

Yang–Mills boundary sources and the Weil form

Mixed identities, source domains, and occurrence obstructions

Drafted for Edward Baker

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Abstract

We investigate whether independently defined sources in a positive four-dimensional Yang–Mills representation can have the complete Weil form as their reflected mixed pairing. A fixed finite $SU(2)$ Wilson slab provides an explicit positive boundary representation. Character packets in its actual interacting state have exact phase branching and limiting winding pairings with the prime-power weights of the explicit formula. A specified compensated radial electric derivative and logarithmic angle insertion recover the full archimedean multiplier, including its contact normalization and logarithmic growth. Together these yield the complete *signed* Weil mixed limit, but not a positive source norm. We prove weak escape, phase incompatibility, bounded-occurrence and closability obstructions, and divergence of positive prime completions. A native current recovers the identity packets strongly with an L^2 pairing; its full spectrum is nevertheless absolutely continuous and cannot carry the required arithmetic correlation. Sharp prime cutoffs have explicit moving boundary responses of norm at least order $\sqrt{A}$. An explicit compact pole-neutral probe detects every nontrivial zeta zero: its translated response is bounded, or tempered, exactly when RH holds. One actual stationary physical realization of that correlation would yield the entire source law by a proved spectral filtering construction. We also give pairing-determination and compact occurrence criteria, and distinguish their unverified arithmetic hypotheses from OS axioms, a mass gap, and phase-space compactness. The physical occurrence and positive completion remain open.

Status. Consolidated research draft; investigation paused at the author’s request. The analytical arguments are model-assisted and have not received independent mathematical review. No proof of RH, of continuum YM existence, or of a YM-to-RH implication is claimed. The model and assumptions are listed in Section 1; provenance and numerical limitations appear in Appendix C.

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1 Scope, arithmetic target, and principal results

The problem is to construct one continuous linear source law

$$J : \mathcal{D}^0 \longrightarrow \mathcal{H}_{\text{YM}}, \quad \langle Jf, Jg \rangle_{\text{OS}} = Q(f, g), \quad (1.1)$$

where the physical model, state, observable prescription and completion are fixed for all compact arithmetic supports. The sources must arise from independently specified YM observables or relations. An arbitrary Hilbert embedding of a form already assumed positive does not solve this problem. Assuming OS positivity, (1.1) implies RH by Weil’s criterion. A nonconstructive source-occurrence theorem is an acceptable route, provided its feasibility hypotheses are independently established.

Most results below concern pure $\text{SU}(2)$ Wilson theory on a spatial 6^3 torus and physical-time slices $-2, -1, 0, 1, 2$, with open outer boundaries, coupling $\beta = 8/5$, and theta angle zero. The boundary state is a finite-slab state, not an identified continuum vacuum. The two-dimensional and thermal calculations in the appendices are explicit comparisons, not changes to this main model.

Full OS reconstruction, continuum YM existence, and a physical mass gap [7] may be assumed where explicitly stated. Local phase-space compactness is a further optional physical assumption. These hypotheses have useful roles in positivity and source compactness, but no argument here derives the missing arithmetic source identity from them. The finite-state weighted electric Laplacian is never identified with the physical continuum Hamiltonian. Arithmetic translation, physical time, contour capacity, and the representation scale are distinct parameters.

1.1 Conventions and the complete form

Let $\mathcal{D} = C_c^\infty(\mathbb{R})$, with its usual compact-support test topology, and use inner products antilinear in the first entry. Put

$$\hat{f}(\tau) = \int_{\mathbb{R}} e^{-i\tau x} f(x) dx, \quad U_t f(x) = f(x - t), \quad \mathbb{T} = -\partial_x^2 + \tfrac{1}{4}. \quad (1.2)$$

The pole-neutral space is

$$\mathcal{D}^0 = \left\{ f \in \mathcal{D} : \int e^{x/2} f(x) dx = \int e^{-x/2} f(x) dx = 0 \right\}. \quad (1.3)$$

For $r > 0$, set

$$n_\Gamma(r) = \frac{e^{-r/2}}{1 - e^{-2r}}, \quad m_+(\tau) = \text{Re } \psi(\tfrac{1}{4} + i\tau/2) - \log \pi, \quad \psi = \Gamma'/\Gamma. \quad (1.4)$$

The full quadratic form is

$$Q[f] = Q_\infty[f] + P[f] - 2 \sum_{a \geq 2} \frac{\Lambda(a)}{\sqrt{a}} \text{Re} \langle f, U_{\log a} f \rangle_2, \quad (1.5)$$

$$Q_\infty[f] = \frac{1}{2\pi} \int_{\mathbb{R}} m_+(\tau) |\hat{f}(\tau)|^2 d\tau, \quad (1.6)$$

$$P[f] = 2 \left| \int f(x) \cosh(x/2) dx \right|^2 - 2 \left| \int f(x) \sinh(x/2) dx \right|^2. \quad (1.7)$$

Here $\Lambda(p^k) = \log p$ and is zero at other integers. Polarization defines $Q(f, g)$. The prime sum is finite for each compact test pair, because sufficiently large shifts have disjoint supports. The pole term vanishes on $\mathcal{D}^0$. On that space Q is stationary and invariant under reflection $f(x) \mapsto f(-x)$.

Equivalently,

$$Q_\infty[f] = m_+(0)\|f\|_2^2 + \frac{1}{2} \iint_{\mathbb{R}^2} |f(x) - f(y)|^2 n_\Gamma(|x - y|) \, dx \, dy. \quad (1.8)$$

The difference makes the diagonal singularity integrable. Its completed contact and asymptotic are

$$m_+(0) = -\gamma_E - \frac{\pi}{2} - 3 \log 2 - \log \pi, \quad m_+(\tau) = \log \frac{|\tau|}{2\pi} + O(|\tau|^{-2}). \quad (1.9)$$

These formulas follow from the digamma integral and asymptotics [5]. In particular, the off-diagonal kernel alone does not fix the required form. The normalization of (1.5) and the sufficiency of positivity on (1.3) are the restricted Weil criterion in [3, Appendices B–C].

Lemma 1.1 (Exact removal of the pole moments). *The map $\mathsf{T} : \mathcal{D} \rightarrow \mathcal{D}^0$ is a topological isomorphism. Its inverse is*

$$(\mathsf{T}^{-1}f)(x) = \int_{\mathbb{R}} e^{-|x-y|/2} f(y) \, dy. \quad (1.10)$$

The inverse preserves an enclosing support interval, although it need not preserve holes in a disconnected support.

Proof. The displayed kernel is the fundamental solution of $-\partial^2 + 1/4$. Its two exterior tails vanish precisely by the two moments. The resulting function is smooth, since its equation is $h'' = h/4 - f$, and is supported in the convex hull of $\text{supp } f$. Integration by parts proves $\mathsf{T}\mathcal{D} \subset \mathcal{D}^0$; compactly supported solutions of $\mathsf{T}h = 0$ vanish. The integral formula and differential equation give continuity on each fixed-support test space, hence the claimed test-space isomorphism. $\square$

Proposition 1.2 (A necessary ultraviolet test). *For nonzero $h \in \mathcal{D}$, the exactly pole-neutral functions*

$$f_\lambda = \lambda^{-2} \mathsf{T}(e^{i\lambda x} h) = e^{i\lambda x} \left(h - \frac{2ih'}{\lambda} + \frac{h/4 - h''}{\lambda^2} \right) \quad (1.11)$$

have uniformly bounded ordinary L^1, L^2, L^∞ norms and satisfy

$$Q[f_\lambda] = \|h\|_2^2 \log |\lambda| + O_h(1) \quad (|\lambda| \rightarrow \infty). \quad (1.12)$$

Consequently no source law locally bounded solely in one of those ordinary input norms realizes Q . In particular this excludes $Jf = \int f(x)A(x) \, dx$ when A is a Hilbert-valued function with locally integrable norm.

Proof. The unmodulated profile in (1.11) converges smoothly to h on a fixed support and has uniform Schwartz Fourier estimates. In (1.6) split at $|\tau - \lambda| = |\lambda|/2$. Equation (1.9) gives the leading logarithm on the first region and the Schwartz tail bounds the second. The finitely many prime terms on the fixed support remain bounded by Cauchy–Schwarz. For the stated integral law, $\|Jf_\lambda\| \leq \|f_\lambda\|_\infty \int_{\text{supp } h} \|A(x)\| \, dx$, contradicting (1.12). $\square$

1.2 What has been established

The affirmative identities in this manuscript are the exact reflected boundary norm, two actual generalized-source constructions with nonarithmetic pairings, the character prime mixed limit, the complete signed archimedean/prime limit, and several sufficient occurrence and determination theorems. The exclusions identify where these identities fail to become a positive Weil-source law. The last criterion reduces the target to one stationary physical correlation, but that correlation is not obtained from a YM observable here. Classical conductor-operator and explicit-formula mathematics is credited to its sources; no claim of novelty is made for those ingredients or for the present deductions without further literature and independent mathematical review.

2 A fixed reflected boundary representation

2.1 Finite slab and reference color fiber

For general $SU(N_c)$ on a finite slab, the Wilson action [1] is

$$S = \beta \sum_p \left(1 - \frac{1}{N_c} \operatorname{Re} \operatorname{tr} U_p \right). \quad (2.1)$$

Use normalized Haar measure on each link. Let U denote the central-slice spatial links and $X_{\pm}$ the strict half-slab variables. Site reflection in physical time gives

$$S(U, X_+, X_-) = S_0(U) + S_+(U, X_+) + S_-(U, X_-), \quad S_- = \theta S_+. \quad (2.2)$$

There is no plaquette containing both strict half interiors. Define

$$h(U) = \int e^{-S_+(U, X_+)} dX_+, \quad \Omega(U) = e^{-S_0(U)/2} h(U), \quad d\nu = Z^{-1} \Omega^2 dm = w dm. \quad (2.3)$$

The compact finite integral shows that Ω is smooth, gauge invariant and strictly positive, with positive lower and upper bounds. This is a prepared finite-slab amplitude with unit outer-boundary data. It is consistent with the standard lattice transfer-matrix framework [2]; the gluing identity needed here is derived directly below.

At a fixed base vertex, retain a reference endomorphism fiber, so a matrix-valued observable transforms by $F \mapsto g F g^{-1}$ there. For a half-slab observable F set

$$\psi_F(U) = e^{-S_0(U)/2} \int e^{-S_+(U, X_+)} F(U, X_+) dX_+. \quad (2.4)$$

Reflected gluing and contraction of the reference indices give

$$\langle F, G \rangle_{\text{refl}} = Z^{-1} \int \frac{1}{N_c} \operatorname{tr}(\psi_F^\dagger \psi_G) dm. \quad (2.5)$$

For boundary functions B, C this reduces exactly to

$$\langle B, C \rangle_\nu = \int \frac{1}{N_c} \operatorname{tr}(B^\dagger C) d\nu, \quad \mathcal{H} = L^2_{\text{cov}}(M, \nu; \operatorname{End} \mathbb{C}^{N_c}). \quad (2.6)$$

The reference fiber retains singlet and adjoint channels and adds no bulk dynamical matter. Tracing each observable before taking its norm would be a different scalar reduction.

Proposition 2.1 (Boundary descent through the reflected null space). *The reflected null space is the kernel of $F \mapsto \psi_F$. For every bounded covariant boundary multiplier D , $\psi_{DF} = D\psi_F$. It therefore descends to the reflected quotient; if $D^\dagger D = I$, its action is unitary. Based spatial loops Q have $\|Q\|_\nu = 1$, and their Gram entries are $\mathbb{E}_\nu \operatorname{tr}(Q_i^\dagger Q_j)/N_c$.*

Proof. Separate the two half integrals in (2.5). Boundary multiplication passes through the half integral, and the remaining statements follow from the positive integral norm. Strict positivity of Ω identifies its completed covariant image with (2.6). $\square$

An interior multiplier need not descend: take independent signs X_+ and X_- with reflected pairing $\mathbb{E} \overline{F(X_-)} G(X_+) = \mathbb{E} \overline{F} \mathbb{E} G$. The function $F(X_+) = X_+$ is null, while multiplication by the unitary sign X_+ sends it to 1. Boundary dependence is the hypothesis that prevents this failure. Slice-only gauge-covariant group-valued smoothing preserves boundary admissibility; its bias and continuum limit still require analysis. General four-dimensional smoothing does not implicitly preserve the same reflection support.

2.2 Weighted derivatives and compact source norms

Return to $SU(2)$. Normalize its bi-invariant metric so $i\sigma_1, i\sigma_2, i\sigma_3$ are orthonormal; the fundamental Casimir is 3. For Haar-divergence-free link derivatives $D_{\ell,j}$,

$$D_{\ell,j}^{*\nu} = -D_{\ell,j} - 2D_{\ell,j} \log \Omega, \quad \mathfrak{e}_\nu(B, C) = \sum_{\ell,j} \langle D_{\ell,j} B, D_{\ell,j} C \rangle_\nu, \quad E_\nu = \sum_{\ell,j} D_{\ell,j}^{*\nu} D_{\ell,j}. \quad (2.7)$$

Take the closed form and its nonnegative Friedrichs operator. The full summed form is gauge invariant, so the covariant subspace reduces it; an individual colored derivative need not preserve that subspace. Ellipticity on compact M implies compact resolvent. For $r > 0$,

$$\mathcal{K}_r = \text{Dom}(1 + E_\nu)^{r/2} \subseteq \mathcal{H}, \quad \|(1 - \mathbf{1}_{[0,L]}(E_\nu))B\| \leq (1 + L)^{-r/2} \|(1 + E_\nu)^{r/2} B\|. \quad (2.8)$$

The inequality follows eigenvalue by eigenvalue and proves compactness by finite-rank approximation. These are physical regularity norms, not arithmetic translation generators.

2.3 The one-plaquette marginal

Fix an elementary spatial plaquette P using $\ell = 4$ distinct links. Its product map onto $SU(2)$ is a smooth submersion: solve for one link with the other three fixed. Disintegration of the smooth positive weight, and differentiation under the compact fiber integral, give

$$P_* \nu = \rho(g) dg, \quad 0 < c_\rho \leq \rho \leq C_\rho < \infty. \quad (2.9)$$

Gauge invariance makes ρ central. Its radial extension is even, smooth and 2π -periodic. The class sector $F(P)I$ is a closed subspace $\mathcal{H}_{\text{cl}}$ of (2.6). Put

$$\chi_n(\theta) = \frac{\sin n\theta}{\sin \theta}, \quad n \geq 1, \quad t_k = \frac{1}{\pi} \int_0^\pi \rho(\theta) \cos(k\theta) d\theta. \quad (2.10)$$

Then $t_{-k} = t_k \in \mathbb{R}$ decreases rapidly, $t_0 - t_2 = 1$, and

$$\langle \chi_n, \chi_m \rangle_\nu = t_{n-m} - t_{n+m}, \quad c_\rho \sum |c_n|^2 \leq \|\sum c_n \chi_n\|_\nu^2 \leq C_\rho \sum |c_n|^2. \quad (2.11)$$

These statements follow from the radial Haar measure $(2/\pi) \sin^2 \theta d\theta$ and the product-to-sum identity. In general $t_0 \neq 1$. The full E_ν need not preserve $\mathcal{H}_{\text{cl}}$; its form compression must be distinguished from a restriction to a reducing subspace.

2.4 What full finite Ward equations determine

For the full finite lattice, suppose a normalized positive measure μ satisfies $\int DF d\mu = \int F(DS) d\mu$ for every smooth F and every link derivative D . Then $\eta = e^S \mu$ satisfies $\int Dh d\eta = 0$, by applying the identity to $F = e^S h$. Hence η is invariant under all link translations and is a constant multiple of product Haar. Thus $\mu = Z^{-1} e^{-S} m$, without a prior density assumption. This determines the state, not an arithmetic source law. A finite list of loop equations does not have the stated all-test hypothesis.

3 A genuine winding/electric source and its limitation

Before passing to character packets it is useful to exhibit a global source law whose admissibility and actual pairing can be proved, and whose failure of the target can also be proved.

Write $w_k = \mathbb{E}_\nu \operatorname{tr}(P^k)/2$; here w_k is a moment, distinct from the full density w in (2.3). Then

$$w_k = t_k - \frac{1}{2}(t_{k+2} + t_{k-2}), \quad q(\theta) = 2\rho(\theta) \sin^2 \theta, \quad (3.1)$$

and for finite Laurent sums

$$\langle \sum a_n P^n, \sum b_m P^m \rangle_\nu = \sum_{n,m} \bar{a}_n b_m w_{m-n}. \quad (3.2)$$

For $f \in \mathcal{D}$, $n \neq 0$, define

$$a_f(n) = |n|^{-1/2} \widehat{f}(\operatorname{sgn}(n) \log |n|), \quad a_f(0) = 0, \quad A_{f,r} = \sum_{n \neq 0} r^{|n|} a_f(n) P^n, \quad 0 < r < 1. \quad (3.3)$$

Every damped observable is smooth. The proposed source is

$$J_E f = \lim_{r \uparrow 1} E_\nu^{1/2} A_{f,r}. \quad (3.4)$$

It is not defined by prescribing a positive arithmetic Gram matrix.

Theorem 3.1 (Admissibility, actual mixed energy, and exclusion). *The limit (3.4) exists in the physical completion, defines a continuous global smooth-test source, and has pairing*

$$\langle J_E f, J_E g \rangle_\nu = \ell \lim_{r \uparrow 1} \sum_{n,m \neq 0} r^{|n|+|m|} \overline{a_f(n)} a_g(m) \{nm w_{m-n} + 2(t_{m-n} - t_{m+n})\}. \quad (3.5)$$

The common Abel limit is essential. There is a constant depending only on the fixed state such that

$$\|J_E f\|_\nu^2 \leq C \ell C_\rho \int (1+x^2) |f(x)|^2 dx. \quad (3.6)$$

Consequently this law, and every fixed scalar multiple of it, fails to realize Q on $\mathcal{D}^0$.

Proof. For $P = \cos \theta I + i \sin \theta \omega \cdot \sigma$, $P^n = \cos(n\theta)I + i \sin(n\theta)\omega \cdot \sigma$. The radial derivative pairing is $nm \cos((m-n)\theta)$, while the angular pairing is $2 \sin(n\theta) \sin(m\theta) / \sin^2 \theta$. Integration gives the braces in (3.5). Each of the four distinct links gives the same bi-invariant Dirichlet contribution, explaining ℓ . This computes the full form; it does not replace E_ν by a radial operator.

For Haar, extend a by zero to $\mathbb{Z}$, put $c(k) = ka(k)$, and use direct Fourier algebra to obtain

$$\|\sum a_n P^n\|_{\text{Haar}}^2 = \frac{1}{2} \sum_k |a(k+2) - a(k)|^2, \quad (3.7)$$

$$\mathfrak{e}_{\text{Haar}}^{(1)}(A, A) = \frac{1}{2} \sum_k |c(k+2) - c(k)|^2 + \sum_k |a(k) - a(-k)|^2. \quad (3.8)$$

For $F = \widehat{f}$, a one-dimensional Sobolev sampling bound gives

$$\sum_{n \geq 1} \frac{|F(\log n)|^2 + |F(-\log n)|^2}{n} \leq C(\|F\|_2^2 + \|F'\|_2^2). \quad (3.9)$$

To prove it, use the fundamental theorem of calculus on $[\log n, \log(n+1)]$, whose length is comparable to $1/n$, then sum. For $c(n) = \sqrt{n}F(\log n)$, the interpolant $b(u) = \sqrt{u}F(\log u)$ satisfies $b'(u) = u^{-1/2}(F'(\log u) + F(\log u)/2)$, so

$$\sum_{n \geq 1} |c(n+2) - c(n)|^2 \leq 4 \int_0^\infty |F'(v) + F(v)/2|^2 dv. \quad (3.10)$$

The negative branch and finitely many crossing indices obey the same bound. This controls (3.8).

For Abel convergence, on the positive branch

$$\Delta_2(r^k c(k)) = r^k \Delta_2 c(k) + (r^2 - 1)r^k c(k+2). \quad (3.11)$$

The first term converges in ℓ^2 . The squared norm of the second is bounded by $C(1-r)^2 \sum_{k \geq 1} (k+2)r^{2k}|F(\log(k+2))|^2$, which tends to zero: remove a finite head and use uniform smallness of F on the tail. The negative branch is identical. Comparison by the positive marginal (2.9) transfers form convergence to the actual state. Closedness proves (3.4) and the mixed identity. Finally Plancherel gives $\|F\|_2^2 + \|F'\|_2^2 = 2\pi \int (1+x^2)|f|^2$, proving (3.6). Proposition 1.2 then excludes the arithmetic match. $\square$

A sharp winding cutoff is not equivalent to Abel convergence: its edge can contain $N|F(\log N)|^2$. The unsmeared coefficient family $|n|^{-1/2 - ix \operatorname{sgn}(n)}$ has a Hilbert interpretation through the first difference norm, but its electric differences behave like $n^{-1/2}$ and fail the form domain. Applying the electric square root only after smearing was necessary, even though the final law remains too locally bounded to be the desired source.

3.1 Euler algebra and the holonomy-twisted Poisson identity

Repeated traversal acts on the continuous functional calculus of P by $\Psi_a F(z) = F(z^a)$. On $C(S^1)$ it is a contraction and $\Psi_a \Psi_b = \Psi_{ab}$. Hence, for $\operatorname{Re} s > 1$,

$$\sum_{a \geq 1} a^{-s} \Psi_a = \prod_p (I - p^{-s} \Psi_p)^{-1}, \quad -Z'(s)Z(s)^{-1} = \sum_{a \geq 2} \Lambda(a) a^{-s} \Psi_a. \quad (3.12)$$

Absolute convergence and the Mobius inverse justify these identities. They are not bounded-operator identities on the actual winding Hilbert norm: the pushed density $q_a(u) = a^{-1} \sum_{j=0}^{a-1} q((u + 2\pi j)/a)$ is positive at $u = \pi$ for every $a \geq 2$, whereas q vanishes quadratically there. A bump of width ϵ at π has input squared norm of order ϵ^3 and output squared norm of order ϵ . Thus Ψ_a has no bounded extension in that metric.

On constants $Z(s)I = \zeta(s)I$, but expectation of a loop is a different operation. Since w_n decreases rapidly, $\sum_{n \geq 1} w_n n^{-s}$ is entire; for Haar it is -2^{-s-1} . There is no scalar zeta Euler product obtained by averaging (3.12).

For a concrete completion test choose $\phi(u) = u^2(2\pi u^2 - 3)e^{-\pi u^2}$. Gaussian differentiation shows that its 2π -Fourier transform equals itself and that $\phi(0) = \int \phi = 0$. The actual observable $\Theta_\phi(t, P) = \sum_{n \in \mathbb{Z}} \phi(nt) P^n$ obeys, at eigenangle $e^{2\pi i \alpha}$,

$$\sum_n \phi(nt) e^{2\pi i n \alpha} = t^{-1} \sum_k \hat{\phi}((k - \alpha)/t). \quad (3.13)$$

The dual lattice is shifted by the holonomy. At $P = I$ alone,

$$\int_0^\infty t^{s-1} \Theta_\phi(t, I) dt = \xi(s)/\pi, \quad \xi(s) = \frac{1}{2} s(s-1) \pi^{-s/2} \Gamma(s/2) \zeta(s). \quad (3.14)$$

Indeed the Mellin transform of ϕ is $s(s-1)\pi^{-s/2}\Gamma(s/2)/(4\pi)$; the sum contributes $2\zeta(s)$. The identity starts in $\operatorname{Re} s > 1$ and continues by the rapid two-ended Poisson decay. Evaluation at $P = I$ has measure-zero support and is not continuous in the physical L^2 norm. The actual mixed pairing is instead $\sum_{n,m} \overline{\phi(nt)}\psi(mu)w_{m-n}$. Thus Euler and Poisson identities alone do not determine a reflected Weil pairing.

4 Character packets, weighted adjoints, and the prime insertion

4.1 Operations in the actual state

On the class sector define

$$(V_a F)(g) = \chi_a(g)F(g^a), \quad V_a \chi_n = \chi_{an}, \quad a \in \mathbb{N}. \quad (4.1)$$

These are independently specified character/winding operations. They satisfy $V_a V_b = V_{ab}$ and $\|V_a\| \leq \sqrt{C_\rho/c_\rho}$, uniformly in a . To see the bound and compute the adjoint, map F to $\sqrt{2}\sin\theta F(\theta)$ in the odd subspace of the circle, with measure $\rho(\theta)d\theta/(2\pi)$. Then V_a becomes $C_a A(\theta) = A(a\theta)$. If

$$L_a h(\theta) = \frac{1}{a} \sum_{j=0}^{a-1} h\left(\frac{\theta + 2\pi j}{a}\right), \quad \rho_a = L_a \rho, \quad (4.2)$$

change of variables gives

$$C_a^{*\rho} = \rho^{-1} L_a M_\rho, \quad V_a^{*\nu} V_a = M_{\rho_a/\rho}. \quad (4.3)$$

The odd subspace is preserved. The familiar divisibility formula $V_a^* \chi_n = \chi_{n/a}$ when $a \mid n$, and zero otherwise, is a *Haar* formula. It cannot be used as the interacting adjoint. Fourier expansion gives $\rho_a \rightarrow t_0$ smoothly as $a \rightarrow \infty$.

4.2 All phase components of a packet

Put $N = e^R$ and, for $\alpha \in \mathbb{R}/(2\pi\mathbb{Z})$, set

$$S_R^\alpha f = \rho(\alpha)^{-1/2} \sum_{n \geq 1} n^{-1/2} e^{in\alpha} f(\log(n/N)) \chi_n, \quad S_R = S_R^0. \quad (4.4)$$

The sum is finite for $f \in \mathcal{D}$. Opposite phases are distinct channels unless $\alpha = 0$ or π .

Theorem 4.1 (Mixed packet limits and exact branching). *For fixed phases and fixed smooth compact tests,*

$$\lim_{R \rightarrow \infty} \langle S_R^\alpha f, S_R^\beta g \rangle_\nu = \begin{cases} \langle f, g \rangle_2, & \alpha = \beta \pmod{2\pi}, \\ 0, & \text{otherwise.} \end{cases} \quad (4.5)$$

For every positive integer a there is the exact identity

$$V_a S_R^\alpha f = \frac{1}{\sqrt{a}} \sum_{a\beta=\alpha \pmod{2\pi}} \sqrt{\frac{\rho(\beta)}{\rho(\alpha)}} S_R^\beta U_{\log a} f. \quad (4.6)$$

In particular,

$$\lim_R \langle S_R f, V_a S_R g \rangle_\nu = a^{-1/2} \langle f, U_{\log a} g \rangle_2, \quad (4.7)$$

$$\lim_R \langle V_a S_R f, V_b S_R g \rangle_\nu = \frac{d \rho_d(0)}{\sqrt{ab} \rho(0)} \langle f, U_{\log(b/a)} g \rangle_2, \quad d = (a, b), \quad (4.8)$$

$$\lim_R \|V_a S_R f\|_\nu^2 = \frac{\rho_a(0)}{\rho(0)} \|f\|_2^2. \quad (4.9)$$

Proof. Insert (2.11) into the packet pairing. For the Toeplitz term fix $k = m - n$. The nonoscillating Riemann sum tends to $e^{ik\alpha} \langle f, g \rangle_2$ when the phases agree, and summing k gives $\sum_k t_k e^{ik\alpha} = \rho(\alpha)$. When the phases differ the extra factor $e^{in(\beta-\alpha)}$ makes the sum tend to zero by summation by parts. The coefficients vary on scale N and are supported between two fixed positive multiples of N . Rapid decay of t_k permits first truncating k , uniformly in R . The Hankel term has $n + m \asymp N$ and tends to zero by that same rapid decay.

For (4.6), sum $e^{im\beta}$ over the a roots of $a\beta = \alpha$. The result is zero unless $a \mid m$, and then is $ae^{i(m/a)\alpha}$. The factors $m^{-1/2}$ and $a^{-1/2}$ give exactly the coefficient of χ_{an} on the left. Equations (4.7)–(4.9) now follow from (4.5): the common roots for a and b are the d roots of zero, whose marginal values sum to $d\rho_d(0)$. $\square$

The extra phase channels account for the full norm in (4.9). Keeping only the identity channel would incorrectly assign the squared norm $a^{-1} \|f\|^2$.

4.3 Euler coefficients and the distributional domain

Uniform boundedness and the semigroup law imply, for $\operatorname{Re} s > 1$,

$$\mathcal{Z}_V(s) = \sum_{a \geq 1} a^{-s} V_a = \prod_p (I - p^{-s} V_p)^{-1}, \quad -\mathcal{Z}'_V(s) \mathcal{Z}_V(s)^{-1} = \sum_{a \geq 2} \Lambda(a) a^{-s} V_a. \quad (4.10)$$

All series and products here converge in operator norm; the absolutely convergent Mobius series is the inverse. These identities explain the coefficient $\Lambda(a)$ without assigning prime weights to a Gram matrix.

At $s = 0$ the same expression is not a Hilbert-space operator on finite characters. In coefficient Sobolev coordinates write $\mathcal{H}_\chi^{-s}$ for the space with norm $\sum_n (1 + n^2)^{-s} |c_n|^2$. For a finite character polynomial F ,

$$\mathcal{L}F = \sum_{a \geq 2} \Lambda(a) V_a F \quad \text{belongs to } \mathcal{H}_\chi^{-s} \quad (s > 1/2). \quad (4.11)$$

Indeed each of its finitely many branches is bounded by $C \log n$, and $\sum n^{-2s} \log^2 n < \infty$. Smooth positive weights give equivalent distributional Sobolev descriptions. In contrast, $\mathcal{L}F \notin \mathcal{H}_{\text{cl}}$ for every nonzero finite character polynomial F . If m is its least nonzero label, then for every sufficiently large prime p the coefficient at mp is exactly $c_m \log p$: another fixed label could divide mp only by dividing m , and those lower coefficients vanish. These coefficients are not square summable.

The legitimate actual mixed matrix entry is nevertheless

$$\langle \chi_n, \mathcal{L}\chi_m \rangle_\nu = \sum_{a \geq 2} \Lambda(a) (t_{n-am} - t_{n+am}), \quad (4.12)$$

which converges absolutely. The notation means distributional duality, not a Hilbert inner product with a purported vector $\mathcal{L}\chi_m$.

Proposition 4.2 (Controlled raw-packet prime tail). *For the raw packets (4.4) at phase zero,*

$$\lim_R \sum_{a \geq 2} \Lambda(a) \{ \langle S_R f, V_a S_R g \rangle_\nu + \langle V_a S_R f, S_R g \rangle_\nu \} = \sum_{a \geq 2} \frac{\Lambda(a)}{\sqrt{a}} \{ \langle f, U_{\log a} g \rangle_2 + \langle f, U_{-\log a} g \rangle_2 \}. \quad (4.13)$$

The sums on the left are absolute distributional pairings. The same limit holds for any cutoff $A(R) \rightarrow \infty$.

Proof. Choose A_0 larger than the ratio of the packet support scales and their inverses. For $a > A_0$, either mixed entry is bounded, for each $M > 0$, by

$$C_{f,g,M} N^{1-M} a^{-M}. \quad (4.14)$$

There are $O(N^2)$ coefficient pairs, their product is $O(N^{-1})$, and $|n - am|$ and $n + am$ are bounded below by a positive multiple of aN in the relevant orientation. Apply the rapid Fourier bound on t in (2.11). Since $\Lambda(a) \leq \log a$, the tail is summable and tends to zero uniformly over cutoffs. For $a \leq A_0$ use (4.7). The limiting correlations are zero beyond another fixed support-dependent cutoff. $\square$

This tail proof concerns raw packets. Later replacement by packets which differ by $o(1)$ in norm preserves each *fixed* bounded V_a pairing, but does not by itself preserve this infinite insertion.

4.4 Weak escape is a physical obstruction

For every nonzero test f , $S_R f \rightarrow 0$ in the actual class sector, while $\|S_R f\| \rightarrow \|f\|_2$. This follows by testing first against finite characters and using (2.11), then density and boundedness. Moreover, for every $r > 0$,

$$\|(1 + E_\nu)^{r/2} S_R f\| \asymp_{f,r} N^r. \quad (4.15)$$

Under Haar, $E_0 \chi_n(P) = \ell(n^2 - 1) \chi_n(P)$. Smooth elliptic Sobolev norm equivalence transfers this estimate to the fixed weighted manifold. Thus the packets violate every fixed positive-order compact-source bound in (2.8). Their limiting mixed numbers cannot be identified with a retained vector merely by weak convergence.

5 The current and a completed signed archimedean response

5.1 The full electric–Wilson current

Let $X = \text{tr}(P)/2 = \cos \theta$ and retain $\ell = 4$. On smooth covariant functions define

$$A_\nu = \frac{i}{2\ell} [E_\nu, X] = -i \left(Y + \frac{1}{2} \text{div}_\nu Y \right), \quad Y = \ell^{-1} \nabla X. \quad (5.1)$$

The complete smooth flow Φ_t of Y on compact M defines the strongly continuous unitary group

$$(\mathcal{U}_t F)(U) = j_t(U)^{1/2} F(\Phi_t U), \quad j_t = \frac{d(\Phi_t^* \nu)}{d\nu}, \quad \mathcal{U}_t = e^{itA_\nu}. \quad (5.2)$$

Its Stone generator fixes the self-adjoint domain; smooth covariant functions form a core. Thus the notation does not depend on selecting an unproved self-adjoint extension of a formal commutator. The actual state correction is

$$A_\nu - A_0 = -\frac{i}{\ell} \nabla \log \Omega \cdot \nabla X. \quad (5.3)$$

It is bounded but generally depends on all links. Replacing it by a class-density drift is not a computation of the full operator.

Proposition 5.1 (The identity phase and the other phases). *With $P_x = -i\partial_x$,*

$$A_\nu S_R^0 f = S_R^0 P_x f + O_{\mathcal{H}}(N^{-1}), \quad (5.4)$$

$$A_\nu S_R^\pi f = S_R^\pi (-P_x f) + O_{\mathcal{H}}(N^{-1}), \quad (5.5)$$

$$A_\nu S_R^\beta f = N \sin \beta S_R^\beta (e^x f) + O_{\mathcal{H}}(1), \quad \beta \neq 0, \pi. \quad (5.6)$$

For $a = 2$ the squared norm of the derivative-intertwining defect tends to $2\rho(\pi)\|f'\|_2^2/\rho(0)$. For $a \geq 3$,

$$\lim_R N^{-2}\|A_\nu V_a S_R f\|^2 = \frac{a}{\rho(0)} \sum_{a\beta=0} \rho(\beta) \sin^2 \beta \int e^{2x} |f(x)|^2 dx > 0 \quad (5.7)$$

when f is nonzero.

Proof. Haar differentiation gives exactly

$$A_0 \chi_n = \frac{i}{2} \left\{ (n + \frac{1}{2}) \chi_{n+1} - (n - \frac{1}{2}) \chi_{n-1} \right\}. \quad (5.8)$$

Taylor-expand the slowly varying coefficients in (4.4). At phase zero the centered difference is $-if'$ and its norm error is $O(N^{-2})$; at phase π the sign reverses. At other phases the leading difference is $n \sin \beta$, giving the displayed N term. For the state drift use its factor ∇X , whose magnitude is $\sqrt{\ell} \sin \theta$. The identity and π packets localize at their respective endpoints, and $\|\sin \theta S_R^{0,\pi} f\| = O(N^{-1})$ by the corresponding coefficient difference estimate. This proves the full weighted errors. Finally apply exact branching and phase orthogonality. For $a=2$ the π derivative differs by $-2P_x$; for larger a some roots have nonzero sine, yielding (5.7). $\square$

More generally a smooth observable $h(X)$ replaces the leading symbol by $Nh'(\cos \beta) \sin \beta$. Finite response at every rational phase requires h' to vanish on a dense subset of $(-1, 1)$, hence h is constant. A positive sum of squared current responses cannot cancel these divergences. At the identity phase alone a current norm produces a derivative-squared response, not the required logarithmic multiplier.

5.2 A specified parity conversion, with its domain

There is nevertheless a signed insertion recovering the complete archimedean term. Its additional compression and compensation must be stated explicitly. Put

$$v(\theta) = \sqrt{\rho(\theta)} \sin \theta, \quad \mathcal{W}_\rho F = \sqrt{2/\pi} v F. \quad (5.9)$$

This is a unitary $\mathcal{H}_{\text{cl}} \rightarrow L^2(0, \pi)$. The compressed electric form divided by ℓ has radial operator $-v^{-2} \partial_\theta (v^2 \partial_\theta)$, and

$$\mathcal{W}_\rho E_{\text{rad}} \mathcal{W}_\rho^{-1} = -\partial_\theta^2 + v''/v. \quad (5.10)$$

The potential extends smoothly to the endpoints; for Haar it is -1 . Subtracting this specified potential leaves the Dirichlet operator $D^2 = -\partial_\theta^2_{\text{Dir}}$. This is a compensated compression, not a restriction of the full E_ν to a reducing class subspace, and no YM Ward identity has been proved to select the subtraction.

The polar derivative

$$B = \partial_\theta D^{-1}, \quad B \sqrt{2/\pi} \sin(n\theta) = \sqrt{2/\pi} \cos(n\theta) \quad (5.11)$$

satisfies $B^* B = I$ and $B B^* = I - \Pi_{\text{constants}}$. Measure the angle in turns, $q = \theta/(2\pi)$, and define

$$C_\rho = \mathcal{W}_\rho^{-1} \{ \log D + B^* M_{\log q} B \} \mathcal{W}_\rho. \quad (5.12)$$

A precise dense symmetric domain is the inverse image of $\{u \in \text{Dom } \log D : (\log q) B u \in L^2(0, \pi)\}$. It contains all finite sine sums, so the finite span of $\phi_n = \rho^{-1/2} \chi_n$ is a common dense test domain for the present matrix calculations. Symmetry implies closability; a positive closed form or self-adjoint realization of the sum is not asserted. The insertion is specified through electric spectral calculus, a polar derivative, and the angle observable. Its multiplier will be derived, not assigned.

5.3 The completed mixed identity

Set

$$F_R f = \sum_{n \geq 1} n^{-1/2} f(\log(n/N)) \phi_n, \quad H_f(t) = t^{-1/2} f(\log t) \quad (t > 0). \quad (5.13)$$

For each test, $\|F_R f - S_R f\|_\nu \rightarrow 0$: the raw sine packet concentrates at $\theta = 0$, where $\rho(\theta)^{-1/2}$ equals $\rho(0)^{-1/2}$. The ϕ_n are an actual orthonormal basis.

Theorem 5.2 (Complete archimedean limit). *For $f, g \in \mathcal{D}$,*

$$\lim_R \langle F_R f, C_\rho F_R g \rangle_\nu = \frac{1}{2\pi} \int m_+(\tau) \widehat{f}(\tau) \widehat{g}(\tau) d\tau. \quad (5.14)$$

This includes the local coefficient and logarithmic growth in (1.9), not only the off-diagonal kernel.

Proof. Let $\mathcal{F}_+ H(y) = 2 \int_0^\infty \cos(2\pi t y) H(t) dt$. It is a unitary involution of $L^2(\mathbb{R}_+)$, by even Fourier extension. Write $u_R = \mathcal{W}_\rho F_R f$. The electric part of the pairing is the Riemann sum

$$R \langle F_R f, F_R g \rangle + \frac{1}{N} \sum_{n \geq 1} \log(n/N) \overline{H_f(n/N)} H_g(n/N).$$

After $\theta = 2\pi y/N$, the cosine part obeys the exact Poisson identity

$$\sqrt{2\pi/N} B u_R(2\pi y/N) = \frac{2}{N} \sum_{n \geq 1} H_f(n/N) \cos(2\pi n y/N) = \sum_{k \in \mathbb{Z}} (\mathcal{F}_+ H_f)(y + kN) \quad (5.15)$$

on $0 < y < N/2$. The even extension of H_f is smooth and supported away from zero, so its transform is Schwartz. All nonzero aliases in (5.15) are uniformly rapidly small on this half interval. The weight $|\log y|$ is integrable near zero, and Schwartz decay controls the expanding tail. As $\log q = \log y - R$, the $-R$ term cancels the electric R term *exactly*, since B is an isometry. The limit is

$$\int_0^\infty \log t \overline{H_f(t)} H_g(t) dt + \int_0^\infty \log y \overline{\mathcal{F}_+ H_f(y)} \mathcal{F}_+ H_g(y) dy. \quad (5.16)$$

This is the real even conductor form; compare [4].

Under the unitary logarithmic coordinate $H(t) = t^{-1/2} f(\log t)$, its Mellin–Fourier conversion is

$$y^{1/2} \widehat{\mathcal{F}_+ H_f}(y)(\tau) = \gamma_+(\tau) \widehat{f}(-\tau), \quad \gamma_+(\tau) = \pi^{i\tau} \frac{\Gamma(1/4 - i\tau/2)}{\Gamma(1/4 + i\tau/2)}.$$

The hat on the left is in $\log y$. One obtains this identity first by the damped cosine Mellin integral $2(2\pi)^{-s} \Gamma(s) \cos(\pi s/2)$ at $s = 1/2 - i\tau$, then by the gamma duplication and reflection formulas. Multiplication by $\log t$ becomes $i\partial_\tau$. In the sum (5.16) the differentiated test terms cancel, while

$$i \gamma'_+(\tau) / \gamma_+(\tau) = \operatorname{Re} \psi(1/4 + i\tau/2) - \log \pi. \quad (5.17)$$

The tests and transformed profiles have the decay needed for integration by parts. The digamma integral and its asymptotic then give (1.8) and (1.9), including the contact. $\square$

Without the parity conversion B , the sine transform gives $\operatorname{Re} \psi(3/4 + i\tau/2) - \log \pi$ and the different kernel $-e^{-3|r|/2}/(1 - e^{-2|r|})$. Rescaling the angle unit adds a local constant. Thus the parity and the 2π Fourier convention are mathematical parts of the construction, not innocuous normalizations to leave unspecified.

For every finite A put

$$K_A = C_\rho - \sum_{2 \leq a \leq A} \Lambda(a)(V_a + V_a^{*\nu}), \quad L_A = m_+(P_x) - \sum_{2 \leq a \leq A} \frac{\Lambda(a)}{\sqrt{a}}(U_{\log a} + U_{-\log a}). \quad (5.18)$$

Its actual finite-core pairing satisfies

$$\lim_{A \rightarrow \infty} \lim_{R \rightarrow \infty} \langle F_R f, K_A F_R g \rangle_\nu = Q(f, g), \quad f, g \in \mathcal{D}^0. \quad (5.19)$$

For fixed A , this follows from Theorem 5.2, (4.7), and the boundedness of each V_a . After taking R the prime sum stabilizes on each compact test pair. No joint cutoff limit for these compensated packets is claimed.

5.4 Why this is not yet a positive source

The insertion (5.12) does not control the phases created by winding. Put

$$F_R^\beta f = \sum n^{-1/2} e^{in\beta} f(\log(n/N)) \phi_n.$$

For $b = \arccos(\cos \beta) \in (0, \pi]$, Fourier localization gives

$$\langle F_R^\beta f, C_\rho F_R^\beta g \rangle_\nu = R \langle f, g \rangle_2 + \langle f, xg \rangle_2 + \log \frac{b}{2\pi} \langle f, g \rangle_2 + o(1). \quad (5.20)$$

Here the electric logarithm still contributes R , whereas the angle packet stays at the nonzero folded angle b , so there is no compensating $-R$. Applying the same localized calculation to $V_a F_R f$ gives

$$\lim_R R^{-1} \langle V_a F_R f, C_\rho V_a F_R f \rangle_\nu = \frac{\sum_{j=1}^{a-1} \rho(2\pi j/a)}{a\rho(0)} \|f\|_2^2 > 0 \quad (a \geq 2). \quad (5.21)$$

Endpoint packets at π have the same conclusion; the cosine localization is interpreted one-sided there.

Even on $\mathcal{D}^0$ the archimedean form alone is not positive. For $h_L = L^{-1/2} h(x/L)$ and $f_L = \mathbb{T} h_L$, its normalized Fourier mass concentrates at zero as $L \rightarrow \infty$. Dominated convergence after scaling gives $Q_\infty[f_L]/\|f_L\|_2^2 \rightarrow m_+(0) < 0$. Thus (5.19) is a proved signed mixed identity. It is neither an OS norm identity nor a proof that subtracting a local divergence preserves positivity.

6 Obstructions to retaining the limiting module

6.1 The profinite comparison and bounded occurrence

The rational phase channels of Theorem 4.1 define the auxiliary limiting Hilbert space

$$\mathcal{K}_\rho = \bigoplus_{\alpha \in 2\pi\mathbb{Q}/(2\pi\mathbb{Z})} L^2(\mathbb{R}), \quad \mathcal{V}_a^\rho(e_\alpha f) = a^{-1/2} \sum_{\alpha\beta=\alpha} \sqrt{\rho(\beta)/\rho(\alpha)} e_\beta U_{\log a} f. \quad (6.1)$$

It records limiting mixed data; it is not declared a new physical sector. Let $D_\rho e_\alpha f = \sqrt{\rho(\alpha)} e_\alpha f$. Then $\mathcal{V}_a^\rho = D_\rho \mathcal{V}_a^1 D_\rho^{-1}$. Fourier series on the profinite integers identify the Haar version with $L^2(\mathbb{R} \times \widehat{\mathbb{Z}})$ and

$$(\mathcal{V}_a^1 F)(x, z) = \sqrt{a} \mathbf{1}_{\widehat{\mathbb{Z}}}(z) F(x - \log a, z/a). \quad (6.2)$$

It is an isometry, with range projection $\mathbf{1}_{\widehat{\mathbb{Z}}}$. The comparison makes a precise occurrence test possible while leaving the YM state fixed.

Theorem 6.1 (No bounded class-sector intertwiner). *Every bounded map $T : \mathcal{K}_\rho \rightarrow \mathcal{H}_{\text{cl}}$ satisfying $T\mathcal{V}_a^\rho = V_a T$ for all a is zero. The conclusion also holds on the closed identity-phase cyclic submodule generated by all $\mathcal{V}_a^\rho e_0 f$.*

Proof. The coefficient map $\mathcal{C} : \ell^2(\mathbb{N}) \rightarrow \mathcal{H}_{\text{cl}}$, $e_n \mapsto \chi_n$, is a bounded isomorphism by (2.11). It intertwines V_a with the shift $S_a e_n = e_{an}$. Thus $A = \mathcal{C}^{-1} T D_\rho$ obeys $A \mathcal{V}_a^1 = S_a A$. For $\eta_n = A^* e_n$ and every prime $p \nmid n$, $(\mathcal{V}_p^1)^* \eta_n = A^* S_p^* e_n = 0$. By the range projection in (6.2), η_n vanishes on $p\mathbb{Z}$ for every such p . The complement of these divisibility sets has Haar measure at most $\prod_{p \in E} (1 - 1/p)$ for each finite set E of primes not dividing n . These products tend to zero by Euler’s divergence of $\sum_p 1/p$. Hence $\eta_n = 0$ and $A = 0$.

For the cyclic submodule, Haar coordinates identify it with the closed span of functions $f(x) \mathbf{1}_{a\mathbb{Z}}(z)$. Multiplication by $\mathbf{1}_{p\mathbb{Z}}$ preserves this subspace because intersections are $\text{lcm}(a, p)\mathbb{Z}$. The same isometry adjoint and range-projection argument applies there. Tests in $\mathcal{D}^0$ are dense in $L^2(\mathbb{R})$: a vector orthogonal to $\mathbb{T}\mathcal{D}$ is an L^2 distributional solution of $\mathbb{T}u = 0$, hence zero. They generate the same cyclic closure. $\square$

In particular, a uniformly bounded family of recovery maps cannot have nonzero strong limits on the whole limiting module while asymptotically preserving all winding relations: its strong limit would be the forbidden bounded intertwiner. The theorem concerns a bounded map into the actual class sector with these specified relations. It is not a no-go theorem for every possible YM source, or for a different physical representation.

6.2 What a fixed closable operation can preserve

Lemma 6.2. *Let T be a fixed densely defined closable operator between Hilbert spaces. If $u_n \in \text{Dom } T$, $u_n \rightharpoonup 0$, and $\sup_n \|Tu_n\| < \infty$, then $Tu_n \rightharpoonup 0$.*

Proof. For every $v \in \text{Dom } T^*$, $\langle v, Tu_n \rangle = \langle T^*v, u_n \rangle \rightarrow 0$. The adjoint domain is dense by closability; boundedness extends the limit to every v . $\square$

Thus a fixed closable insertion cannot recover a nonzero strong vector from the raw escaping packets while keeping their output norms bounded. This does not forbid a scale-dependent unitary recentring, which will succeed for the ordinary L^2 pairing.

There is a second, global obstruction even after that recovery.

Theorem 6.3 (No closable global L^2 completion of the Weil norm). *Let $J_0 : L^2(\mathbb{R}) \rightarrow \mathcal{H}$ be any isometry. There is no fixed closable operator T with $J_0 \mathcal{D}^0 \subset \text{Dom } T$ such that*

$$\langle TJ_0 f, TJ_0 g \rangle = Q(f, g) \quad (f, g \in \mathcal{D}^0). \quad (6.3)$$

Proof. The assumed identity first implies positivity and hence RH by Weil’s criterion. This is a consequence in a contradiction proof, not an extra hypothesis. Under that consequence the explicit formula reads

$$Q(f, g) = \sum_{\gamma \text{ distinct}} m_\gamma \overline{\widehat{f}(\gamma)} \widehat{g}(\gamma). \quad (6.4)$$

Fix one ordinate γ_0 and $h \in \mathcal{D}$ with $\widehat{h}(0) = 1$. Set

$$f_L = \frac{\mathbb{T}}{\gamma_0^2 + 1/4} \left[L^{-1} e^{i\gamma_0 x} h(x/L) \right]. \quad (6.5)$$

Then $\|f_L\|_2 \rightarrow 0$, while

$$\widehat{f}_L(\gamma) = \frac{\gamma^2 + 1/4}{\gamma_0^2 + 1/4} \widehat{h}(L(\gamma - \gamma_0))$$

converges in $\ell^2(\{m_\gamma\})$ to the unit point vector at γ_0 . To justify the norm convergence, use isolation of γ_0 , the Schwartz estimates of $\widehat{h}$, and $N(T) = O(T \log T)$; a fixed summable majorant controls all other ordinates for $L \geq 1$. Hence TJ_0f_L is Cauchy with nonzero limiting squared norm m_{γ_0} , even though $J_0f_L \rightarrow 0$. This contradicts closability. $\square$

The support of f_L grows. The conclusion concerns closability relative to the ordinary global L^2 input norm; it does not preclude continuous distributional sources on fixed-support smooth-test spaces, or closable physical operators in a stronger source topology.

7 Strong recentering and the full current spectrum

7.1 A retained source with ordinary L^2 pairing

For the complete vector field in (5.1),

$$Y\theta = -\sin\theta, \quad \tan(\theta_t/2) = e^{-t} \tan(\theta/2), \quad \operatorname{div}_m Y = -3 \cos\theta. \quad (7.1)$$

These follow from the four distinct link derivatives of P. Consequently

$$j_t(U) = \left(\frac{\sin\theta_t}{\sin\theta} \right)^3 \frac{w(\Phi_t U)}{w(U)}. \quad (7.2)$$

For $P \neq -I$ the trajectory converges as $t \rightarrow +\infty$: its remaining length is $\theta_t/\sqrt{\ell}$, since $\|Y\| = \sin\theta/\sqrt{\ell}$. Write $\Phi_\infty U$ for this limit and $w_\infty(U) = w(\Phi_\infty U)$. This is a bounded positive measurable function. Smoothness at the repelling critical set is not asserted.

Theorem 7.1 (Native strong recovery). *The actual vectors $J_R f = \mathcal{U}_R S_R f$ converge strongly to*

$$J_{\text{rec}} f(U) = \sqrt{\frac{w_\infty(U)}{\rho(0)w(U)}} J_{\text{H}} f(\theta(U)) I, \quad (7.3)$$

$$J_{\text{H}} f(\theta) = \frac{(1 + y^2/4)^{3/2}}{y} \int_0^\infty H_f(t) \sin(yt) dt, \quad y = 2 \tan(\theta/2). \quad (7.4)$$

Moreover

$$\langle J_{\text{rec}} f, J_{\text{rec}} g \rangle_\nu = \langle f, g \rangle_2, \quad J_{\text{rec}} U_t = \mathcal{U}_{-t} J_{\text{rec}}. \quad (7.5)$$

Thus J_{rec} extends to an isometry from $L^2(\mathbb{R})$ into the same physical boundary completion.

Proof. First use Haar. The map

$$(\mathcal{R}F)(y) = \sqrt{2/\pi} \frac{y}{(1 + y^2/4)^{3/2}} F(2 \arctan(y/2)) \quad (7.6)$$

is a unitary from radial Haar L^2 to $L^2(\mathbb{R}_+, dy)$. Apply it to the Haar recentered packet and use (7.2). The exact result is

$$\frac{\sqrt{2/\pi}}{N \sqrt{1 + (y/(2N))^2}} \sum_{n \geq 1} H_f(n/N) \sin(2n \arctan(y/(2N))). \quad (7.7)$$

On bounded y intervals the Riemann sums converge uniformly to $\sqrt{2/\pi} \int H_f(t) \sin(yt) dt$. The squared norm before this limit is exactly $N^{-1} \sum |H_f(n/N)|^2 \rightarrow \|H_f\|_2^2$. The sine transform is unitary with this normalization, so local convergence, boundedness, and equality of limiting norms give strong convergence on the whole half-line. Inverting $\mathcal{R}$ gives J_H .

For the actual measure, $\mathcal{U}_R S_R f$ equals the Haar expression times $\sqrt{w(\Phi_R U)/(\rho(0)w(U))}$. These multipliers are uniformly bounded and converge almost everywhere to the factor in (7.3). Strong convergence follows by dominated convergence and the Haar strong limit. Unitarity of $\mathcal{U}_R$ and (4.5) prove the pairing. Finally $S_R U_t f = S_{R+t} f$ exactly. Passing to the strong limit in $\mathcal{U}_R S_{R+t} f = \mathcal{U}_{-t}(\mathcal{U}_{R+t} S_{R+t} f)$ proves covariance. $\square$

The formula describes actual full-state dependence through w_∞/w . It does not assume that the current preserves the class sector. Since $\mathcal{U}_R$ is unitary, $\mathcal{U}_R F_R f$ has the same strong limit.

Winding retains only part of its norm under this recentering:

$$\mathcal{U}_R V_a S_R f \rightharpoonup a^{-1/2} J_{\text{rec}} U_{\log a} f, \quad (7.8)$$

while the missing limiting squared norm is

$$\left(\frac{\rho_a(0)}{\rho(0)} - \frac{1}{a} \right) \|f\|_2^2 = \frac{\sum_{j=1}^{a-1} \rho(2\pi j/a)}{a\rho(0)} \|f\|_2^2 > 0 \quad (a \geq 2). \quad (7.9)$$

Indeed each nonidentity branch, after the flow, tends to zero on compact sets away from $P = -I$ by summation by parts in its oscillatory coefficient series. Boundedness and density give weak convergence to zero there. Its retained norm escapes toward the repelling set, which has measure zero. Equations (4.6) and (4.9) then prove (7.8)–(7.9). This recentering solves weak escape for the unbranched L^2 source, but does not retain the full winding module or the Weil norm.

7.2 An exact normal form for the entire current

Theorem 7.2 (Full interacting current is absolutely continuous). *There is a transverse covariant Hilbert space $\mathcal{K}$ and a unitary identification*

$$\mathcal{H} \simeq L^2(\mathbb{R}, dr; \mathcal{K}), \quad \mathcal{U}_t H(r) = H(r-t), \quad A_\nu = i\partial_r, \quad (7.10)$$

with the usual Hilbert-valued H^1 generator domain. In particular every finite-vector coefficient satisfies

$$\langle \xi, \mathcal{U}_t \eta \rangle_\nu \longrightarrow 0 \quad (|t| \rightarrow \infty). \quad (7.11)$$

Proof. Off $P = \pm I$, put $r = \log \tan(\theta/2)$, so $Yr = -1$. Every remaining trajectory crosses the regular hypersurface $\Sigma = \{\theta = \pi/2\}$ once. The map $(r, z) \mapsto \Phi_{-r} z$ is a smooth bijection with smooth inverse from $\mathbb{R} \times \Sigma$ onto this full-measure part of M . The excluded sets have Haar measure zero by the plaquette submersion, and therefore have ν measure zero. This coordinate statement includes every link, not just the plaquette angle.

Choose a smooth positive gauge-invariant transverse measure σ , and write the pulled-back measure as $b_\nu(r, z) dr d\sigma(z)$. Multiplication by $b_\nu^{1/2}$ changes it to product measure. The weighted flow factor is exactly $\sqrt{b_\nu(r-t, z)/b_\nu(r, z)}$, so this multiplication converts the group into translation. Gauge transformations preserve r and commute with the flow, restricting only the transverse Hilbert space. Fourier transformation in r proves pure absolute continuity. Each matrix coefficient is the Fourier transform of an L^1 function, by Hilbert-valued Cauchy–Schwarz, and hence tends to zero. $\square$

On the recovered subspace (7.5) implies $A_\nu J_{\text{rec}} = J_{\text{rec}} P_x$ on the smooth-test domain. The full theorem is stronger than that subspace statement: changing to an arbitrary seed vector elsewhere in the physical boundary space cannot turn this current into a pure point arithmetic translation generator.

8 Positive completions and the moving prime boundary

8.1 The unsubtracted positive prime form has zero domain

For finite A define the positive expression

$$q_A^{\text{prime}}[F] = \sum_{2 \leq a \leq A} \Lambda(a) \|(I - V_a)F\|_\nu^2. \quad (8.1)$$

Its negative cross terms resemble the prime part of Q, but its diagonal terms cannot be omitted from a positive identity.

Proposition 8.1. *In the full actual class sector,*

$$\left\{ F : \sum_{a \geq 2} \Lambda(a) \|(I - V_a)F\|_\nu^2 < \infty \right\} = \{0\}. \quad (8.2)$$

Proof. Coefficient shifts and the norm equivalence (2.11) give $V_a F \rightarrow 0$ as $a \rightarrow \infty$, first for finite characters and then for all F by uniform boundedness. Equation (4.3) gives

$$\|V_a F\|_\nu^2 = \int |F|^2 \rho_a \, dg \rightarrow t_0 \|F\|_{\text{Haar}}^2.$$

Consequently $\|(I - V_a)F\|_\nu^2 \rightarrow \|F\|_\nu^2 + t_0 \|F\|_{\text{Haar}}^2$, which is strictly positive for nonzero F. The sum already diverges along the infinitely many primes. $\square$

On raw packets the limiting positive form at finite A is

$$\left[\sum_{2 \leq a \leq A} \Lambda(a) \left(1 + \frac{\rho_a(0)}{\rho(0)} \right) \right] \|f\|_2^2 - 2 \sum_{2 \leq a \leq A} \frac{\Lambda(a)}{\sqrt{a}} \text{Re} \langle f, U_{\log a} f \rangle_2. \quad (8.3)$$

The bracket diverges while, for a fixed compact f, the cross sum stabilizes. Pole neutrality does not remove the bracket. Formal subtraction leaves a signed expression; positivity of the unrenormalized forms supplies no proof of positivity of that difference.

8.2 A coercive variational attempt also diverges

Let h be any densely defined closed quadratic form on the class space with $h[u] \geq \delta \|u\|_\nu^2$ for $\delta > 0$. Set

$$q_A[u] = h[u] + q_A^{\text{prime}}[u], \quad K_A^{\text{pos}} \geq \delta I. \quad (8.4)$$

Each q_A is closed on Dom h. For a bounded readout $B : \mathcal{H}_{\text{cl}} \rightarrow \mathcal{Y}$ define

$$\Gamma_A[y] = \inf_{Bu=y} q_A[u]. \quad (8.5)$$

When feasible, the affine constraint is closed and strict coercivity gives a unique minimizer. The vector $(K_A^{\text{pos}})^{1/2} u_A$ is an actual positive source. For finite-dimensional surjective B,

$$u_A = (K_A^{\text{pos}})^{-1} B^* [B(K_A^{\text{pos}})^{-1} B^*]^{-1} y. \quad (8.6)$$

Theorem 8.2 (Failure of a fixed bounded constrained readout). *For every nonzero feasible y , $\Gamma_A[y] \rightarrow +\infty$. Also $(K_A^{\text{pos}})^{-1} \rightarrow 0$ and $(K_A^{\text{pos}})^{-1/2} \rightarrow 0$ strongly.*

Proof. If a subsequence of constrained energies were bounded, coercivity would give $u_{A_j} \rightarrow u$ along a subsequence. For each fixed finite cutoff B , lower semicontinuity of its bounded positive prime form gives $q_B^{\text{prime}}[u] \leq C$. Taking $B \rightarrow \infty$ and using Proposition 8.1 forces $u=0$, whereas weak continuity of the readout forces $Bu=y$. This is a contradiction.

For $u_A = (K_A^{\text{pos}})^{-1}v$, coercivity bounds both u_A and $q_A[u_A] = \langle u_A, v \rangle$. Every weak limit is zero by the same argument without a readout. Thus $q_A[u_A] \rightarrow 0$ and $\|u_A\| \rightarrow 0$. Finally $\|(K_A^{\text{pos}})^{-1/2}v\|^2 = \langle v, u_A \rangle \rightarrow 0$. $\square$

For the constraint that the Haar character coefficient $c_m = b$, one has the explicit lower bound

$$q_A[u] \geq c_\rho |b|^2 \left[\delta + \sum_{\substack{p \leq A \\ p \text{ prime}}} \frac{\delta \log p}{\delta + \log p} \right]. \quad (8.7)$$

Indeed keep the mass terms at m and the distinct labels pm , and the p -th difference at pm . Minimizing $\delta |c_{pm}|^2 + \log p |c_{pm} - b|^2$ gives the summand. A physical mass gap can motivate coercivity only after an actual excited-sector identification. Even granting it here does not repair this divergent positive construction.

8.3 An explicit boundary layer of the signed prime operator

The sharp arithmetic cutoff has an additional, independently provable norm obstruction. Let $h \in \mathcal{D}$ be real, supported in $[-1, 1]$, and put $f = \mathbb{T}h$. Define

$$\mathcal{P}_A f = \sum_{2 \leq a \leq A} \frac{\Lambda(a)}{\sqrt{a}} (U_{\log a} + U_{-\log a}) f, \quad L = \log A. \quad (8.8)$$

Theorem 8.3 (Prime-cutoff edge profiles). *Uniformly for $s \in [-1, 1]$,*

$$A^{-1/2} \mathcal{P}_A f(L + s) \rightarrow h'(s) + \tfrac{1}{2} h(s), \quad (8.9)$$

$$A^{-1/2} \mathcal{P}_A f(-L + s) \rightarrow -h'(s) + \tfrac{1}{2} h(s). \quad (8.10)$$

For the completed signed operator L_A in (5.18),

$$\liminf_{A \rightarrow \infty} A^{-1} \|L_A f\|_2^2 \geq 2 \|h'\|_2^2 + \tfrac{1}{2} \|h\|_2^2 > 0 \quad (8.11)$$

if h is nonzero. This uses the prime number theorem, not RH.

Proof. Let $\Psi(x) = \sum_{a \leq x} \Lambda(a)$. At the positive edge only one shift orientation contributes for large A , and its scaled Stieltjes sum is

$$\int_0^1 q^{-1/2} f(s - \log q) d\{\Psi(Aq)/A\}.$$

The tests vanish unless $q \geq e^{-2}$. The prime number theorem $\Psi(x) = x + o(x)$, uniformly on this fixed positive q interval after division by A , and integration by parts give convergence to the Lebesgue

integral, uniformly in s . Uniformity follows from bounded smooth derivatives in the compact (q, s) rectangle. The integral equals

$$e^{s/2} \int_s^1 e^{-u/2} f(u) du = h'(s) + h(s)/2,$$

because $f = -h'' + h/4$ and all endpoint derivatives vanish. The negative edge similarly gives $e^{-s/2} \int_{-1}^s e^{u/2} f(u) du = -h'(s) + h(s)/2$.

Outside the support of f , the archimedean operator has kernel $-n_\Gamma(|x - y|)$. Expanding it into exponentials shows that its $e^{-|x|/2}$ tail vanishes by the pole moments; the remaining tail is $O(e^{-5|x|/2})$. It is negligible at both moving edges after scaling. Integrate the two squared edge profiles on their disjoint intervals. Their cross derivative terms cancel, leaving $2\|h'\|^2 + \|h\|^2/2$. $\square$

Any counterterm whose output stays in a fixed compact neighborhood of the original support, even with A -dependent coefficients, misses these moving intervals and cannot cancel (8.11). The theorem does not cover every smooth regulator, or a nonlocal moving counterterm with an independently justified physical definition.

This is also a lower bound in the actual state, not solely an arithmetic calculation. For finite A let $q_{R,A} = \mathcal{U}_R K_A F_R f$, with the signed K_A of (5.18). Strong recentering and its finite- A mixed identity imply

$$\liminf_R \|q_{R,A}\|_\nu \geq \|L_A f\|_2. \quad (8.12)$$

To prove this, take any subsequence of bounded norms and a weak limit q . Pair with the strongly convergent $\mathcal{U}_R F_R g \rightarrow J_{\text{rec}} g$. The limit is $\langle g, L_A f \rangle_2$. Density of $\mathcal{D}$ and the isometry of J_{rec} force the projection of q onto its range to be $J_{\text{rec}} L_A f$. If the norms are unbounded there is nothing to prove. Combining (8.12) and (8.11) gives an iterated physical norm lower bound of order $\sqrt{A}$. No joint cutoff convergence is inferred.

9 Determination and sufficient source-occurrence theorems

9.1 Off-diagonal data leave exactly three local constants

Suppose B is a continuous Hermitian form on $\mathcal{D}^0$, stationary under U_t and invariant under reflection. Assume its mixed pairing on $\mathbb{T}h, \mathbb{T}k$ agrees with Q whenever h and k have disjoint supports. This hypothesis includes the prime shifts and the archimedean off-diagonal kernel, with their differentiated pullbacks.

Theorem 9.1 (Three-anchor determination). *If also $B[f_\lambda] = O_h(\log |\lambda|)$ for the normalized modulations (1.11), then*

$$B(\mathbb{T}h, \mathbb{T}k) - Q(\mathbb{T}h, \mathbb{T}k) = a_0 \langle h, k \rangle_2 + a_1 \langle h', k' \rangle_2 + a_2 \langle h'', k'' \rangle_2 \quad (9.1)$$

for three real constants. Equality on three anchors $\mathbb{T}h_r$, where $h_r(x) = r^{-1/2}h(x/r)$, any nonzero $h \in \mathcal{D}$, and $r = 1, 2, 3$, forces $B = Q$ everywhere.

Proof. Pull the difference back to $\mathcal{D}$ through the topological isomorphism $\mathbb{T}$. The distribution kernel theorem and stationarity give a convolution kernel. Disjoint-support vanishing makes that kernel supported at zero, hence it is a finite linear combination of derivatives of δ . Hermiticity makes its Fourier polynomial real; reflection makes it even. For $f_\lambda = \lambda^{-2} \mathbb{T}(e^{i\lambda x} h)$, a term of degree d has leading size $\lambda^{d-4} \|h\|^2$. The logarithmic bound and Proposition 1.2 therefore exclude $d > 4$. This proves (9.1). At the three scaled anchors the coefficients multiply $\|h\|^2, r^{-2}\|h'\|^2, r^{-4}\|h''\|^2$. The determinant is $-5\|h\|^2\|h'\|^2\|h''\|^2/54$, which is nonzero. $\square$

Three constants are necessary in this argument. Without reflection the Hermitian polynomial may have every degree from zero through four, leaving five real constants. Symmetry, growth, and finitely many anchors *without* the off-diagonal identity are inadequate: a real even Schwartz multiplier can be chosen to vanish on all those finitely many anchor functionals yet change the form.

One useful necessary test is visible after pullback. At each nonzero prime shift the kernel of $Q(\mathsf{T}h, \mathsf{T}k)$ contains

$$-\frac{\Lambda(a)}{\sqrt{a}} \mathsf{T}_r^2 \{\delta(r - \log a) + \delta(r + \log a)\}, \quad \mathsf{T}_r^2 = \partial_r^4 - \frac{1}{2} \partial_r^2 + \frac{1}{16}. \quad (9.2)$$

A candidate with a smooth off-diagonal mixed kernel cannot supply these singularities. Negative off-diagonal coefficients alone are not an obstruction to positivity: squared shift differences supply such terms together with positive contact.

9.2 Compactness turns verified finite feasibility into occurrence

Theorem 9.2 (Compact-source occurrence criterion). *Let $\mathcal{K} \subseteq \mathcal{H}_{\text{phys}}$ be an injectively and compactly embedded Hilbert source space. Choose a countable complex-rational linear test core in $\mathcal{D}^0$, dense on every enclosing compact support interval. Prescribe for each coordinate f a bound $\|Jf\|_{\mathcal{K}} \leq C_I P_I(f)$, using fixed continuous test seminorms on each interval I . Suppose every finite collection of the desired source relations is feasible by actual vectors in these same balls. The relations must be closed in the strong physical topology and include complex-rational linearity and the required mixed pairings on the test core. Then there is a continuous global source law satisfying all these relations and the same bounds.*

Proof. Each $\mathcal{K}$ ball has compact image in $\mathcal{H}_{\text{phys}}$. Its image is closed: a strong physical limit of bounded $\mathcal{K}$ vectors has a weak $\mathcal{K}$ subsequence, whose image equals that strong limit. Take the product of these compact coordinate balls. Every imposed closed relation cuts out a closed subset. Finite feasibility is exactly the finite intersection property, so a common assignment exists. For each support interval the seminorm bound extends the linear assignment from the dense core to the full test space. Rational complex linearity and continuity give complex linearity. Compatible approximations in larger intervals give one global law; physical strong continuity passes all mixed pairings to the limit. Weak compactness in $\mathcal{K}$ preserves the stated source bounds. $\square$

Unbounded operator relations require their actual closed graph and sufficient coordinate bounds, rather than a formal symbolic identity. A high-frequency estimate needed for Theorem 9.1 must likewise be imposed uniformly by a common continuous seminorm; asymptotic agreement on a countable list does not imply it. A useful special case is a sequence of full source maps with uniform compact-source bounds and convergent mixed data on a dense core; a diagonal compactness argument then gives the same conclusion.

The finite boundary model supplies the spaces $\mathcal{K}_r$ in (2.8), but the raw character packets fail their bounds. More fundamentally, feasibility of the arithmetic relations with these bounds has not been established. Positivity of separately assembled finite matrices does not establish the theorem’s physical finite-feasibility hypothesis.

9.3 Optional continuum OS and phase-space hypotheses

Assume a reconstructed vacuum theory with local algebras $\mathcal{A}(O)$, vacuum Ω_{vac} , and physical Hamiltonian $H_{\text{phys}} \geq 0$. A mass gap means $H_{\text{phys}} \geq \Delta(1 - P_{\text{vac}})$. It controls infrared excitations, but does not imply compactness of arbitrary ultraviolet source families.

A natural additional phase-space hypothesis is that, for bounded O and $\beta > 0$,

$$\Theta_{\beta,O} : \mathcal{A}(O) \longrightarrow \mathcal{H}_{\text{OS}}, \quad A \longmapsto e^{-\beta H_{\text{phys}}} A \Omega_{\text{vac}} \quad (9.3)$$

maps the operator-norm unit ball into a relatively compact set. Nuclearity is a familiar stronger condition [8]; it is an explicit extra assumption here, not a consequence claimed from the mass gap.

For each finite M the set

$$\mathcal{K}(O, M) = \{e^{-\beta H_{\text{phys}}} A \Omega_{\text{vac}} : A \in \mathcal{A}(O), \|A\| \leq M\} \quad (9.4)$$

is actually compact. Relative compactness is assumed. For closedness, take a norm-convergent sequence of images, an ultraweak convergent subnet of their bounded operators, and test against arbitrary vectors. The vector functionals are normal, so the norm limit is the image of the ultraweak limit operator in $\mathcal{A}(O)$ with the same bound. There is also the elementary energy-tail estimate

$$\|\mathbf{1}_{[E,\infty)}(H_{\text{phys}})e^{-\beta H_{\text{phys}}} A \Omega_{\text{vac}}\| \leq M e^{-\beta E}. \quad (9.5)$$

Consequently the preceding finite-intersection proof applies with these compact prepared-vector sets. If every finite set of required arithmetic/source relations is realizable with fixed local regions O_I and fixed operator bounds $M_f \leq C_{IP_I}(f)$, it produces a global actual source $Jf = e^{-\beta H_{\text{phys}}} A_f \Omega_{\text{vac}}$ with those relations. If the vacuum is separating for the relevant local algebra, injectivity makes A_f unique and transfers vector linearity to operator linearity; otherwise only the vector law is needed. Reeh–Schlieder density by itself does not provide the uniform bounds or arithmetic feasibility. No such feasibility is proved here, even when all OS axioms, a mass gap, and phase-space compactness are granted.

10 A single detecting correlation and an exact occurrence reduction

10.1 A compact test which misses no nontrivial zero

Let $a_j = 2^{-j}$ and let b be the density of the absolutely convergent sum of independent uniforms on $[-a_j, a_j]$. Its bilateral Laplace transform is

$$\mathcal{B}(z) = \prod_{j \geq 1} \frac{\sinh(2^{-j} z)}{2^{-j} z}. \quad (10.1)$$

Removable factors at zero have value one. Define

$$h_* = \frac{e^x b(x)}{\mathcal{B}(1)}, \quad f_* = \mathbb{T} h_*, \quad F(z) = \int e^{zx} f_*(x) dx = (1/4 - z^2) \frac{\mathcal{B}(1+z)}{\mathcal{B}(1)}. \quad (10.2)$$

Lemma 10.1. *The real functions h_*, f_* are smooth, supported in $[-1, 1]$, and $f_* \in \mathcal{D}^0$. The entire function F has no zeros in $|\operatorname{Re} z| < 1/2$. In particular $\widehat{f_*}(\tau) \neq 0$ at every real τ .*

Proof. The random sum is supported in $[-1, 1]$. On the imaginary axis, bound any fixed number k of product factors by $\min(1, (a_j |\tau|)^{-1})$ and the rest by one. Its Fourier transform decays faster than every power, giving a smooth compact density by Fourier inversion. The product converges locally uniformly in the plane because its tail deviations from one are summable. All zeros of each factor lie on the imaginary axis, and off that axis the nonzero summable tail cannot vanish. For $|\operatorname{Re} z| < 1/2$, one has $\operatorname{Re}(1+z) > 1/2$, so neither the product nor $1/4 - z^2$ vanishes. Integration by parts proves (10.2) and the two neutral moments. $\square$

The tilt e^x removes the real Fourier zeros that an untilted even compact density would have. This probe uses no zero list and no assumption about RH.

Put $k(t) = \langle f_*, U_t f_* \rangle_2$ and define the real even response

$$C_*(t) = Q(f_*, U_t f_*) = \frac{1}{2\pi} \int m_+(\tau) |\widehat{f}_*(\tau)|^2 e^{-it\tau} d\tau - \sum_{a \geq 2} \frac{\Lambda(a)}{\sqrt{a}} \{k(t + \log a) + k(t - \log a)\}. \quad (10.3)$$

Here k is smooth and supported in $[-2, 2]$, so the prime sum is locally finite in t . The definition contains no spectral zero data. For $t > 2$ its archimedean part is

$$C_{*,\infty}(t) = - \sum_{j \geq 1} e^{-(2j+1/2)t} F(2j+1/2) F(-2j-1/2). \quad (10.4)$$

This follows by expanding the off-diagonal kernel. The $j = 0$ term vanishes by pole neutrality, and the local contact contributes only for overlapping supports.

10.2 An unconditional criterion and an unconditional nondecay theorem

To avoid confusion with the plaquette density ρ , denote a nontrivial zeta zero by ϱ , counted with multiplicity, and put $\lambda_\varrho = \varrho - 1/2$. The unconditioned explicit formula [3] gives

$$C_*(t) = \sum_{\varrho} c(\lambda_\varrho) e^{\lambda_\varrho t}, \quad c(\lambda) = F(\lambda) F(-\lambda). \quad (10.5)$$

Opposite Fourier conventions are reconciled by $\varrho \mapsto 1 - \varrho$. Off the critical line the coefficient is this product, not a squared modulus. It is nonzero at every nontrivial zero by Lemma 10.1. The zero symmetries, $0 < \operatorname{Re} \varrho < 1$, and $N(T) = O(T \log T)$ are unconditional [5, 6]. Smooth compact support gives rapid vertical decay of F uniformly in bounded real strips. Thus the sum converges absolutely and locally uniformly with every t derivative.

Theorem 10.2 (One-probe regularity criterion). *The following are equivalent: RH; boundedness of C_* on $\mathbb{R}$; temperedness of the distribution defined by C_* ; and, for every $\epsilon > 0$, a bound $|C_*(t)| \leq C_\epsilon e^{\epsilon t}$ for $t \geq 0$. Moreover*

$$\limsup_{t \rightarrow +\infty} \frac{\log(1 + |C_*(t)|)}{t} = \sup_{\varrho} |\operatorname{Re} \varrho - 1/2|. \quad (10.6)$$

Proof. Under RH the expansion becomes

$$C_*(t) = \sum_{\gamma \text{ distinct}} m_\gamma |\widehat{f}_*(\gamma)|^2 e^{-i\gamma t}. \quad (10.7)$$

The positive coefficients are summable; this proves all three regularity properties.

Conversely, in $\operatorname{Re} z > 1/2$ absolute convergence permits the one-sided Laplace transform

$$\int_0^\infty e^{-zt} C_*(t) dt = \sum_{\varrho} \frac{c(\lambda_\varrho)}{z - \lambda_\varrho}. \quad (10.8)$$

The right side is meromorphic on the plane, normally convergent off its discrete poles; the residue at a zero of multiplicity m is $mc(\lambda) \neq 0$. A subexponential bound makes the left side holomorphic on $\operatorname{Re} z > 0$. Temperedness has the same consequence: multiply the smooth function distribution by a

smooth cutoff equal to one for $t \geq 1$ and zero for $t \leq 0$, take its exponentially damped distributional Laplace transform, and add the compact initial-interval correction. This avoids multiplying an arbitrary distribution by a discontinuous step. The transform agrees with the integral in its initial half-plane and is holomorphic for $\operatorname{Re} z > 0$. It cannot have one of the nonzero residues there. The functional equation excludes left-half-plane λ_ϱ as well. Boundedness is a special case.

For (10.6) let b be the supremum on the right, also the supremum of $\operatorname{Re} \lambda_\varrho$. Absolute summability bounds $|C_*(t)|$ by $e^{bt} \sum |c(\lambda_\varrho)|$. If the limsup were less than b , a slightly larger exponent still below b would give a holomorphic Laplace transform to the left of at least one nonzero-residue pole, a contradiction. For $b=0$ the upper bound and the nonnegative logarithm give equality directly. $\square$

This is a relation between RH and a precise regularity property, not a derivation of that property from OS temperedness. The variable t is logarithmic character scale, and the response has not been shown to be a Schwinger function in spacetime. Simply postulating its temperedness would postulate an RH-equivalent arithmetic condition without establishing a physical explanation.

Corollary 10.3. *Unconditionally, $C_* \notin C_0(\mathbb{R})$.*

Proof. Membership in C_0 would imply boundedness and hence RH. Under that consequence, the absolutely convergent series (10.7) is uniformly almost periodic and

$$\lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T |C_*(t)|^2 dt = \sum_{\gamma \text{ distinct}} \{m_\gamma |\hat{f}_*(\gamma)|^2\}^2 > 0.$$

One proves the formula first for finite sums and passes by uniform convergence. There are nontrivial zeros and the probe misses none. A function in C_0 has zero such mean square, a contradiction. $\square$

Together with Theorem 7.2, this excludes $C_*(t) = \langle \xi, \mathcal{U}_t \eta \rangle$ for *any* two physical Hilbert vectors. In particular no current-covariant source law with the exact Weil pairing can exist in the fixed boundary space, even if its seed is not a class function or a finite-order generalized source. An inserted coefficient $\langle v, K \mathcal{U}_t v \rangle$ is also excluded whenever $v \in \operatorname{Dom} K^*$, because it is the ordinary coefficient $\langle K^* v, \mathcal{U}_t v \rangle$. A genuinely distributional left functional is outside this finite-vector statement and is not an OS norm.

Similarly, a diagonal cutoff $A(R)$ cannot produce bounded actual response vectors with all translated mixed limits equal to C_* . A weakly convergent subsequence of such bounded vectors, paired with the strongly recovered translates in (7.5), would give a finite-vector current coefficient equal to C_* , contradicting the corollary. This is a conditional obstruction to that proposed joint limit, not a claim that the joint convergence holds.

10.3 One actual stationary correlation would suffice

Theorem 10.4 (One-correlation source occurrence). *Suppose an independently specified physical closed admissible sector contains a vector v and a strongly continuous unitary group $V_t = e^{-itG}$ such that*

$$\langle v, V_t v \rangle_{\text{phys}} = C_*(t) \quad (t \in \mathbb{R}). \quad (10.9)$$

Then RH follows, and spectral filtering in that same sector constructs a continuous global source $J : \mathcal{D}^0 \rightarrow \mathcal{H}_{\text{phys}}$ with $Jf_ = v$, $JU_t = V_t J$, and $\langle Jf, Jg \rangle_{\text{phys}} = Q(f, g)$.*

Proof. The left side of (10.9) is bounded by $\|v\|^2$, so Theorem 10.2 gives RH. Fourier uniqueness for finite measures and (10.7) determine the actual spectral measure of v :

$$d\mu_v(\lambda) = \sum_{\gamma \text{ distinct}} m_\gamma |\hat{f}_*(\gamma)|^2 \delta_\gamma(d\lambda). \quad (10.10)$$

For $g \in \mathcal{D}^0$ define

$$Jg = \left(\frac{\hat{g}}{\hat{f}_*} \right) (G)v. \quad (10.11)$$

There are no real denominator zeros, but no polynomial bound for this quotient is presumed. Its actual functional-calculus domain is verified instead:

$$\int \left| \frac{\hat{g}(\lambda)}{\hat{f}_*(\lambda)} \right|^2 d\mu_v = \sum_{\gamma} m_{\gamma} |\hat{g}(\gamma)|^2 < \infty.$$

Thus bounded spectral truncations converge to an actual vector in the same closed sector. Polarization gives (6.4), hence Q by the explicit formula. The ratio for $g = f_*$ is one; multiplication by $e^{-it\lambda}$ proves covariance. Finally for tests in a fixed compact interval, integration by parts bounds $|\hat{g}(\gamma)|$ by $C_{I,M}(1 + |\gamma|)^{-M}$ times a fixed finite derivative seminorm. Zero counting makes the squared sum finite, proving test-space continuity. $\square$

An equivalent hypothesis is an actual norm-continuous stationary family $\{v_t\}$ with $\langle v_s, v_t \rangle = C_*(t - s)$: translation on its linear span is an isometry, extends to a unitary group on its closure, and gives (10.9). A separately identified native arithmetic generator is therefore optional. No family with this correlation has been constructed here. The theorem guarantees vectors in the declared closed sector, not bounded local operators or stronger electric source bounds unless those additional domains are independently preserved.

11 Status, corrections, and the remaining gap

The strongest constructive fixed-state result is the complete signed mixed identity (5.19), with the actual weighted adjoints, a controlled raw prime distribution, and the full archimedean contact and logarithmic growth. Its proof uses smooth positive finite-slab weights, elementary compact-group analysis, Fourier/Mellin identities, and the stated iterated limit. It does not use RH. Strong recentering independently produces a genuine source with ordinary L^2 norm, but that norm is not Q.

The strongest current-specific obstruction is Theorem 7.2 together with Corollary 10.3: the full weighted electric–Wilson current has only absolutely continuous spectrum and cannot carry even the single detecting response between finite vectors. The prime-boundary lower bound and the positive zero-domain theorem explain two distinct failures of naive completion. Neither a fixed closable L^2 completion nor the tested coercive positive readout removes them.

The audits require the following corrections or qualifications to informal earlier interpretations.

- Use the weighted adjoint (4.3); the Haar divisibility rule is not the interacting adjoint.
- Keep all winding phases. The phase π carries the opposite current; other nonidentity phases have the growth in (5.7).
- The infinite Euler insertion is distributional on finite characters, not a Hilbert vector. Its raw-packet tail theorem does not justify a joint limit after an unbounded change of insertion or packet.
- The archimedean insertion is a specified compensated radial compression. Its classical conductor calculation is exact, but a physical principle selecting the compensation and angle normalization has not been established.

- Weak packet convergence does not retain norms. Scale-dependent native recentering repairs the identity-phase L^2 source and loses the extra winding norm in the precise amount (7.9).
- A divergent positive local coefficient cannot be deleted while citing positivity of the original form. The completed signed form requires a separate positive realization or positivity proof.
- Full OS axioms, a mass gap, and optional local phase-space compactness have the roles stated in Section 9; none supplies the unproved arithmetic finite-feasibility or correlation identity. A native generator is an optional ansatz, not a prerequisite imposed by Weil positivity.

The exact remaining mathematical gap is to prove *physical occurrence*: either independently specified admissible YM sources with the full pairing (1.1), finite arithmetic feasibility with the compact preparation bounds of Theorem 9.2, or the one actual stationary correlation in Theorem 10.4. The last theorem reduces the amount of mixed data that needs to be established, but its hypothesis is substantial and unverified. It is not legitimate to define the physical spectral measure by the zero ordinates and then invoke that theorem.

The investigation is paused at the author’s request. This draft consolidates the completed calculations, conditional theorems, scoped exclusions, and diagnostics; it does not extend them into a proof of RH or an implication from four-dimensional YM existence and mass gap. All new deductions remain model-assisted work awaiting independent mathematical review.

A Wilson-loop hierarchies and the interacting experiment

A.1 Continuous storage, memory, and projection error

This appendix records the physical hierarchy from which the source investigation developed. Its continuous statements assume a positive regulated measure on smooth Hermitian connections and bounded multiplication generators with integrable operator norm on the capacity interval. They are not asserted for unrenormalized continuum loops.

Let $q(t) \in \mathbb{R}^2$ be the tip of a planar trace starting at the origin. Close the trace by its straight return chord. If U_t is transport along the trace and $R_t(r)$ along the chord from 0 to $rq(t)$, define

$$Q_t = R_t(1)^{-1}U_t, \quad J_t = \int_0^1 r \hat{F}_{12,t}(r) dr, \quad k_t = q_1 \dot{q}_2 - q_2 \dot{q}_1. \quad (\text{A.1})$$

The hat denotes curvature transported back along the chord. Variation of parallel transport, or differentiation of its path-ordered equation with the endpoint terms cancelled, gives $\dot{Q}_t = ik_t J_t Q_t$. The moment J_t is Hermitian and traceless. Put $A_t X = ik_t J_t X$ in the positive trace-normalized configuration Hilbert space, $e = I$, $\Psi_t = Q_t$, and $W(t) = \langle e, \Psi_t \rangle$. The generator is skew-adjoint and the propagator is unitary. For $P = |e\rangle\langle e|$ and $z = (I - P)\Psi$,

$$|W|^2 + \|z\|^2 = 1, \quad \dot{W} = -\langle v_t, z_t \rangle, \quad \dot{z}_t = v_t W + (I - P)A_t(I - P)z_t, \quad v_t = A_t e. \quad (\text{A.2})$$

The first identity is orthogonal decomposition. It describes storage, not monotone dissipation: a two-dimensional unitary rotation returns all amplitude. It also concerns one prepared state, not arbitrary-input passivity.

The compressed bounded skew-adjoint generator has a unitary propagator $R(t, s)$. Eliminating z with $z_0 = 0$ gives the exact memory equation

$$\dot{W}(t) = - \int_0^t \Gamma(t, s) W(s) ds, \quad \Gamma(t, s) = \langle v_t, R(t, s) v_s \rangle. \quad (\text{A.3})$$

If $w_t = R(t, 0)^* v_t$, then $\Gamma(t, s) = \langle w_t, w_s \rangle$ for both time orders, so the full two-time kernel is positive definite and $\|z_T\|^2 = \|\int_0^T w_s W(s) ds\|^2$. Writing $g(t) = \mathbb{E} \text{tr}(J_t^2)/N_c$ also gives $|\Gamma(t, s)| \leq |k_t k_s| \sqrt{g(t)g(s)}$. The causally truncated kernel is not thereby a positive self-adjoint operator, and no stationarity in capacity has been proved. This is an explicit time-dependent projection identity in the spirit of [9]; evaluating the complementary propagator can remain as difficult as the original dynamics.

For a differentiable linearly independent family $B(t)c = \sum_j c_j b_j(t)$ containing e , let $G = B^* B$, $M = B^* A B$, $C = B^* \dot{B}$. Galerkin orthogonality gives

$$G\dot{c} = (M - C)c, \quad \frac{d}{dt}(c^* G c) = 0. \quad (\text{A.4})$$

Indeed $M^* = -M$ and $\dot{G} = C + C^*$; omitting the moving connection C invalidates conservation. For $D = \dot{B} - AB$ and $\Pi = BG^{-1}B^*$, the actual residual is

$$r = (I - \Pi)Dc, \quad \|r\|^2 = c^*(D^* D - D^* B G^{-1} B^* D)c. \quad (\text{A.5})$$

This is a nonnegative Gram Schur complement. Singular families must first be quotiented by their null space. If the initial vector is represented exactly, Duhamel's formula and unitarity imply

$$\|\Psi_t - B(t)c(t)\| \leq \int_0^t \|r_s\| ds, \quad (\text{A.6})$$

$$|W(t) - \langle e, B(t)c(t) \rangle| \leq \int_0^t \left(\int_s^t |k_v| \sqrt{g(v)} dv \right) \|r_s\| ds. \quad (\text{A.7})$$

For the second inequality use $\langle e, r_s \rangle = 0$ and subtract e from the backward-propagated readout. Its difference has norm at most the inner integral. Initial error adds its norm to both bounds.

For the two-observable family $e, J/\sqrt{g}$, assume $g > 0$ and put $\mu_j = \mathbb{E} \text{tr}(J^j)/N_c$ and $d = \mathbb{E} \text{tr}(J^2)/N_c$. The internal moving connection vanishes and

$$\frac{d}{dt} \begin{pmatrix} x \\ y \end{pmatrix} = ik \begin{pmatrix} 0 & \sqrt{g} \\ \sqrt{g} & \mu_3/g \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}, \quad (x(0), y(0)) = (1, 0). \quad (\text{A.8})$$

The orthogonal derivative and curvature residuals are $R_D = \dot{J} - (\dot{g}/(2g))J$ and $R_C = J^2 - ge - (\mu_3/g)J$. Their inner product is real, so the cross term in $\|R_D - ikR_C\|^2$ vanishes:

$$\|r_2\|^2 = \frac{|y|^2}{g} \left[d - \frac{\dot{g}^2}{4g} + k^2 \left(\mu_4 - g^2 - \frac{\mu_3^2}{g} \right) \right]. \quad (\text{A.9})$$

Both summands in brackets are nonnegative squared orthogonal residuals. The displayed plus sign is essential; it repairs a missing addition sign in the preliminary note's displayed equation (5.8). For $SU(2)$, $\mu_3 = 0$, so $x = \cos \vartheta$, $y = i \sin \vartheta$, with $\vartheta = \int_0^t k_s \sqrt{g(s)} ds$. Although J^2 is scalar configuration by configuration, its coefficient fluctuates, and $\mu_4 - g^2$ need not vanish.

For the linear driver $u(t) = at$, the regulated tip expansion gives $k_t = -(2a/3)\sqrt{t} + O(t^{3/2})$ and $J_t = F_{12}(0)/2 + O(\sqrt{t})$. If $C_0 = \mathbb{E} \text{tr}(F_{12}(0)^2)/N_c > 0$, then

$$\vartheta = -\frac{2a\sqrt{C_0}}{9}t^{3/2} + O(t^2), \quad W_2 = 1 - \frac{2a^2 C_0}{81}t^3 + O(t^{7/2}).$$

Under $\|R_D\| = O(t^{-1/2})$ and $\|R_C\| = O(1)$, the refined output bound is $O(t^{7/2})$, while the whole-state bound is only $O(t^2)$. One sufficient regulated set of moment hypotheses is finite $\mathbb{E}K_1^2$ and

$\mathbb{E}K_0^4$, where K_0 bounds curvature and K_1 its directional covariant derivative in the swept region. Chord transport and the ordered commutator term give

$$g \leq \frac{1}{4}\mathbb{E}K_0^2, \quad \mu_4 \leq \frac{1}{16}\mathbb{E}K_0^4, \quad d \leq \mathbb{E} \left(|\dot{q}|K_1/3 + |k|K_0^2/4 \right)^2. \quad (\text{A.10})$$

The factors follow from $\int_0^1 r \, dr = 1/2$, $\int_0^1 r^2 \, dr = 1/3$, and the ordered-commutator bound using $\|J(r)\| \leq K_0 r^2/2$. No regulator-uniform version is established.

A.2 Exact discrete contour steps

For actual lattice loops Q_n , multiplication by $U_n = Q_{n+1}Q_n^*$ is unitary and maps Q_n to Q_{n+1} . It uses no fractional link powers. Relative to e and its orthogonal complement write the blocks as a_n, b_n, c_n, d_n . With initial complement zero,

$$W_{n+1} = a_n W_n + \sum_{j < n} b_n d_{n-1} \cdots d_{j+1} c_j W_j. \quad (\text{A.11})$$

This is exact block elimination; resetting the complement would change the evolution.

For moving finite families B_n containing e , set $G_n = B_n^* B_n$ and $T_n = B_{n+1}^* U_n B_n$. Orthogonal projection gives

$$c_{n+1} = G_{n+1}^{-1} T_n c_n, \quad \rho_n^2 = \|B_{n+1} c_{n+1} - U_n B_n c_n\|^2 = c_n^* (G_n - T_n^* G_{n+1}^{-1} T_n) c_n. \quad (\text{A.12})$$

Here retained squared norm decreases by exactly ρ_n^2 ; this discrete compression loss differs from the continuous Galerkin conservation above. Telescoping unitary propagation gives whole-state error at most $\sum_{j < N} \rho_j$. Writing $K_{Nj} = \langle Q_N, Q_j \rangle$, a sharper scalar bound is

$$|W_N - \widehat{W}_N| \leq \sum_{j < N} \eta_{Nj} \rho_j, \quad \eta_{Nj}^2 = 2 - 2 \operatorname{Re} K_{N,j+1}, \quad (\text{A.13})$$

where residual j lives at step $j+1$. The factor is at most $\min(2, \sum_{k=j+1}^{N-1} \sqrt{2 - 2 \operatorname{Re} K_{k+1,k}})$. These follow by subtracting e from the exact backward readout at that step, which is orthogonal to the residual.

A computable backward family gives another bound. Start with $\chi_N = e = B_N h_N$ and recurse $h_j = G_j^{-1} T_j^* h_{j+1}$. Its backward residual has norm

$$\delta_j^2 = h_{j+1}^* (G_{j+1} - T_j G_j^{-1} T_j^*) h_{j+1}. \quad (\text{A.14})$$

Backward telescoping bounds the error in the readout at step $j+1$ by $\sum_{k=j+1}^{N-1} \delta_k$. Since the forward residual is orthogonal to that retained readout, one obtains

$$|W_N - \widehat{W}_N| \leq \sum_{j < N} \rho_j \sum_{k=j+1}^{N-1} \delta_k. \quad (\text{A.15})$$

All are inequalities for true ensemble Gram entries. Floating Monte Carlo evaluation does not turn them into certified ensemble bounds.

A.3 The first interacting finite-slab test

The completed experiment used the fixed $SU(2)$ 6^3 by 5 open-time Wilson lattice at $\beta = 1.6$. Geometry was specified before sampling: linear driver $u(t) = t$, capacities 0, 1, 2.8, 6, nearest-vertex trace prefixes and straight return chords. The signed lattice areas were 0, -1 , -2 , -6 . The continuum tip obeys $\dot{q} = 1 - 2/q$ and $q = 2i\sqrt{t} + (2/3)t - (i/18)t^{3/2} + t^2/135 + \dots$. Integration in $\sqrt{t}$ used maximum step 10^{-4} , started at 10^{-5} . Halving the mesh left all words unchanged; the independent equation $q + 2\log(1 - q/2) = t$ had residual below $3.8 \cdot 10^{-12}$, with tip refinement differences below $3.4 \cdot 10^{-12}$.

Four seeded chains, two cold and two Haar starts, used 600 burn-in sweeps and 2400 production sweeps each, recording every sixth sweep: 1600 total configurations. Each sweep used three proposals per link, with uniformly distributed Pauli axis and angle in $[-1.2, 1.2]$; acceptance was approximately 0.599. The 216 spatially translated loops were averaged within each configuration, not counted as independent samples. Correlated batch bootstrap used 400 resamples and block lengths 5, 10, 20, and 40; the principal quoted choice was 20. Estimated autocorrelation scales of 0.50–0.76 measurement intervals do not prove mixing.

The measured plaquette was 0.385525 ± 0.000207 . The loop expectations were

Capacity	Wilson mean	Estimated standard error
1	0.388006	0.000716
2.8	0.150381	0.000939
6	0.002404	0.000891

The tested local family used an anti-Hermitian transported plaquette-curvature average B with $\|B\|_{\text{op}} \leq 1/2$. The retained families were I ; I, B ; and $I, B, v^2 I$, where $B^2 = -v^2 I$ in $SU(2)$. The unknown completed loop was not included as a basis vector. At capacity 2.8 the results were

Dimension	Predicted mean	Difference from direct	Output bound
1	0.150548	$+0.000168 \pm 0.000746$	0.849452
2	0.135442	-0.014939 ± 0.000666	0.800524
3	0.136709	-0.013671 ± 0.000678	0.798915

The first-step residual norms were approximately 0.92166, 0.86983, 0.86949; retained squared norms were 0.15055, 0.24340, 0.24398. Most state norm was lost. The near scalar coincidence for dimension one is not evidence of factorization, and larger nested families need not improve a later scalar readout. These families did not establish useful closure.

Earlier prescribed-background controls were more favorable: for a noncommuting three-background $SU(2)$ ensemble with weights 0.2, 0.3, 0.5, slope 0.8 and capacity 0.4225, the direct loop was 0.9996561469, the two-mode result 0.9996420564, and the refined output bound about $8.00 \cdot 10^{-5}$ versus observed error $1.41 \cdot 10^{-5}$. The whole-state bound was about 0.00707. This demonstrates the possible value of the projection bound in a controlled ensemble, not its efficacy in YM. The ensemble was not reflection/parity invariant.

The sampler sources, geometry rules, seeds, checks, and small record are retained in `numerics/`. Raw chains and executables are regenerable scratch outputs, not a permanent deposited data archive. No uncertainty estimate quoted here is an interval certificate, a continuum extrapolation, or evidence for an arithmetic source identity.

B Source domains, spectral tests, and model comparisons

B.1 Completion is not a physical source prescription

On the same compact boundary manifold take two smooth positive gauge-invariant weights w_0, w_1 . Then

$$U_{01}F = \sqrt{w_0/w_1} F \quad (\text{B.1})$$

is unitary between their covariant L^2 spaces and intertwines every bounded boundary multiplication. Smooth positive multiplication and elliptic Sobolev equivalence also make it a bounded isomorphism of every fixed positive-order source space $\mathcal{K}_r$. Both assertions follow directly by cancellation in the norm and bounded smooth multiplier estimates. Thus bare Hilbert occurrence, covariance, and local regularity bounds do not distinguish these Wilson weights. The distinguished vector is changed: $U_{01}I = \sqrt{w_0/w_1}I$. Relations with that vector or a fixed electric operator are generally not preserved unless conjugated. Nor is an exact finite-word loop algebra preserved by arbitrary smooth weight multiplication. This is why an independently specified state-dependent source rule is essential.

For clarity, the countable core in Theorem 9.2 can be made explicit: choose a dense sequence in each rational-endpoint compact support interval, include the three anchors, and close over $\mathbb{Q}(i)$ under rational translations, reflection, and rational modulations e^{iqx} . Apply $\mathbf{T}$ to that profile space. Partitions of unity for separated supports are applied to the profiles h, k , not to the neutral tests Th, Tk , since arbitrary multiplication need not preserve neutrality. Continuity then extends rational translations and modulations to real ones.

For an already continuous source J , pointwise $\|J[\lambda^{-2}\mathbf{T}(e^{i\lambda x}h)]\|^2 = O_h(\log(2 + |\lambda|))$ for *every* h on a fixed support interval implies a common seminorm bound. Divide these continuous linear maps by $\sqrt{\log(2 + |\lambda|)}$ and apply the uniform boundedness principle on the Fréchet test space. Continuity in λ controls bounded parameter intervals. This reasoning does not apply when the pointwise premise is known only on a dense list.

Membership in $\mathcal{K}_r$ alone gives Sobolev regularity, not a bounded or smooth observable. If $d = \dim M$, $r > d/2$ suffices for continuous bounded representatives; smoothness needs bounds of all orders or a separate smoothing statement. L^2 completion suffices for mixed norms and bounded multiplications, but arbitrary products, finite loop polynomials, and unbounded insertions have separate domain obligations.

B.2 The generator induced by an exact pairing

If an exact source J exists, stationarity of Q makes $Jf \mapsto JU_t f$ an isometry on its range. It is well defined on null vectors and extends to a strongly continuous unitary group on the source closure. No independently named YM generator is required for this conclusion.

Positivity then implies RH, and (6.4) identifies the completed source space with $\ell^2(\{\gamma\}, m_\gamma)$ through Fourier evaluation. Its image is dense: a sequence orthogonal to all $\hat{f}(\gamma)$, $f = Th$, defines a tempered point-supported spectral distribution by Cauchy–Schwarz and zero counting. Its inverse Fourier distribution annihilates $\mathbf{T}\mathcal{D}$; multiplication by $\tau^2 + 1/4$ then forces the original distribution to vanish. Translation has one eigenline per distinct ordinate, with m_γ a norm weight. Multiplicity of a zero does not require a Jordan block or m_γ independent eigenvectors.

This generator is unbounded in both spectral directions and cannot be periodic. One can check these requirements even before invoking the exact zero model: the modulations of Proposition 1.2 obey $Q[f'_\lambda]/Q[f_\lambda] \sim \lambda^2$, and reflection sends the generator to its negative. If U_P acted identically for some $P > 0$, choose h in a support shorter than P . The neutral difference $f_\lambda - U_P f_\lambda$ would have zero source norm, whereas its disjoint leading profiles give $Q[f_\lambda - U_P f_\lambda] = 2\|h\|^2 \log |\lambda| + O(1)$, a

contradiction. Finite-dimensional generators and fixed periodic color rotations therefore cannot realize it.

Nor can arithmetic translation be identified with $e^{-tH_{\text{phys}}}$ for a nonnegative Euclidean Hamiltonian on the source vectors. Exact norm stationarity for all positive t forces every source into $\ker H_{\text{phys}}$ by the spectral theorem. Then $JU_t f = Jf$, and differentiation gives $Jf' = 0$, contradicting the high-frequency test. This excludes that ansatz, not general use of a physical Hamiltonian.

A well-defined generalized spectral source associated with a self-adjoint G and vector η is

$$Jg = \{\widehat{g}(\lambda)(1 + \lambda^2)^{s/2}\}(G)\eta, \quad \langle Jf, Jg \rangle = \int \overline{\widehat{f}(\lambda)} \widehat{g}(\lambda)(1 + \lambda^2)^s d\mu_\eta(\lambda). \quad (\text{B.2})$$

For smooth compact tests the combined multiplier is bounded, with fixed-support test seminorm bounds. It must not be factored into an unbounded operation on η unless that separate domain is verified. Multiplication by this spectral weight cannot turn an absolutely continuous spectral measure into the required atoms. For a complete weighted boundary flow, multiplication by $\sqrt{w}$ conjugates the half-divergence generator to the corresponding Haar generator; a smooth density alone does not create an arithmetic spectrum.

B.3 Modular, boost, and thermal controls

The boundary matrix multiplication algebra has the faithful finite trace $\tau(B) = \int \text{tr}(B) d\nu/N_c$. In its trace GNS space, $S(B) = B^*$ is antiunitary, so the modular operator is $\Delta = I$. A nonconstant central density does not supply a nontrivial modular Hamiltonian in this algebra.

For a proposed continuum wedge alternative, assume the positive-energy Poincaré representation, a unique vacuum, and the applicable wedge modular identification with boosts [11]. A boost eigenvector has a finite joint momentum spectral measure invariant under that boost: the phase of the eigenvector cancels. On $p_\pm = p_0 \pm p_1 > 0$, the logarithms of these coordinates translate under boosts. A finite translation-invariant measure on $\mathbb{R}$ must vanish, by disjoint translates of an interval. Thus the momentum measure is supported where $p_+ = p_- = 0$. The forward-cone condition forces the full momentum to be zero, and uniqueness of the vacuum leaves only the vacuum vector. Hence nonzero arithmetic eigenvalues cannot occur as ordinary wedge-boost eigenvectors under these hypotheses. No mass gap is needed for this argument. The wedge identification for a YM net is itself conditional; it does not follow from the finite slab.

A thermal Liouville comparison changes the state. For a discrete Hamiltonian $H|n\rangle = E_n|n\rangle$ with finite Gibbs partition function, put $r_n = Z^{-1}e^{-\beta E_n}$ and $L = H_{\text{left}} - H_{\text{right}}$ on Hilbert–Schmidt operators. For bounded self-adjoint A and $v = A\rho_\beta^{1/2}$, direct matrix expansion gives

$$d\mu_v(\lambda) = \sum_{m,n} r_n |A_{mn}|^2 \delta_{E_m - E_n}(d\lambda), \quad \mu_v(-d\lambda) = e^{-\beta\lambda} \mu_v(d\lambda). \quad (\text{B.3})$$

This detailed balance law is incompatible with an even measure away from zero. The centered choice $v_c = \rho_\beta^{1/4} A \rho_\beta^{1/4}$ gives instead weights $\sqrt{r_m r_n} |A_{mn}|^2$, which are even. Schatten Hölder gives $\|v_c\|_{\text{HS}} \leq \|A\|$. Consequently ordinary orbit smearing is still locally bounded and fails Proposition 1.2. An independently justified generalized source would need its actual spectrum and domains analyzed. In the vacuum, $[H, A]\Omega_{\text{vac}} = HA\Omega_{\text{vac}}$; this is not a two-sided thermal Liouville spectrum.

Arithmetic quotient representations such as Meyer’s [10] supply a different comparison. Their virtual character is poles minus zeros, so its sign on pole-neutral tests is opposite the desired zero contribution. Derivative-evaluation jets occur in the continuous dual and are not automatically Hilbert vectors. If a proposal insists on realizing all such jets faithfully with a positive norm and unitary translation, a nontrivial nilpotent Jordan block is impossible: its exponential has

polynomially growing orbits. That ansatz would also require simple zeros. The Weil norm does not require simplicity, since it uses multiplicities as positive weights.

B.4 What topology and the two-dimensional comparison do and do not change

The winding/electric local bound used a finite compact link manifold, smooth positive state, and a simple contour with distinct links, not its contractibility. Changing to another such finite graph or a simple noncontractible cycle preserves the same type of obstruction. More generally, a limit of pairings $B_\alpha[f] \leq C_I \|f\|_2^2$ with a common fixed-support constant obeys the same bound. Finite sums, and Hilbert direct sums with summable squared bounds, do not evade it. A continuum or singular-state limit could help only if that bound loses uniformity, while existence and positivity of the new source still need proof.

The bare character Mellin law also fails this test:

$$J_\chi f = \sum_{n \geq 1} n^{-1/2} \widehat{f}(\log n) \chi_n, \quad \langle J_\chi f, J_\chi g \rangle = \sum_{n, m} \frac{\widehat{f}(\log n) \widehat{g}(\log m)}{\sqrt{nm}} G_{nm}.$$

The coefficient form is bounded by $C \int (1 + x^2) |f(x)|^2 dx$ by logarithmic sampling and (2.11). Character dressing improves the winding operator algebra, but not this bare law.

For comparison only, two-dimensional SU(2) YM has Haar character basis and, in the chosen Casimir units, cylinder and handle operators

$$P_t \chi_n = e^{-t(n^2-1)} \chi_n, \quad H_{\text{hand}} \chi_n = n^{-2} \chi_n, \quad \mu(\chi_n \otimes \chi_m) = \delta_{nm} n^{-1} \chi_n. \quad (\text{B.4})$$

These are the usual character gluing formulas [12, Proposition 5.7]. On Haar packets they give exactly

$$N^{2j} H_{\text{hand}}^j S_R f = S_R(e^{-2jx} f), \quad P_{t/N^2} S_R f = e^{t/N^2} S_R(e^{-te^{2x}} f). \quad (\text{B.5})$$

Thus the rescaled actions tested here are local multipliers, while fixed positive area or a fixed positive number of handles kills the packets. The logarithm $L_{\text{hand}} = -\frac{1}{2} \log H_{\text{hand}}$ satisfies $L_{\text{hand}} V_a = V_a(L_{\text{hand}} + \log a)$ and $(L_{\text{hand}} - R) S_R f = S_R(xf)$; it supplies the integer scale, not arithmetic differentiation. For genus $g \geq 2$, the zero-area partition function is $\sum_n n^{2-2g} = \zeta(2g-2)$. This special-value identity is not a mixed Weil norm. SO(3) keeps odd dimensions and changes the dimension Dirichlet series to $(1-2^{-s})\zeta(s)$. These are changes of dimension or gauge group, not results in the fixed four-dimensional state.

Finally the modular-surface scattering coefficient is $C_{\text{sc}}(s) = \Lambda_\xi(2s-1)/\Lambda_\xi(2s)$, where $\Lambda_\xi(s) = \pi^{-s/2} \Gamma(s/2) \zeta(s)$ [13, Section 7.3.2]. The entire completion $\xi(s) = s(s-1)\Lambda_\xi(s)/2$ gives the algebraic relation

$$\frac{\xi(z)}{\xi(z+1)} = \frac{z-1}{z+1} C_{\text{sc}}((z+1)/2). \quad (\text{B.6})$$

This is one meromorphic transfer-function comparison at shift 1/2, not a YM preparation/evolution/readout law. The separate project target $K_\omega(z) = \xi(1/2 + z - \omega)/\xi(1/2 + z + \omega)$ and its ordinary causal energy balance require additional physical identities beyond a reflected Weil pairing. Arithmetic thermal partition functions, adelic trace formulas, or proposed arithmetic cohomological pairings likewise do not by themselves supply those identities.

C Research provenance, diagnostics, and reproducibility

C.1 Scope of this consolidation

This manuscript consolidates the YM investigation as it stood on 25 September 2026. It introduces no new numerical experiment or physical candidate. The historical notes and reviews remain in place; the manuscript is one maintained document, not a new dated snapshot directory. The baseline checkout was

f8b8622fbd687bd152dd40dcc0c4bbd4576008e5.

That commit identifies the starting checkout; it does not claim to contain the later uncommitted notes or this draft. The build record fingerprints the actual research files used, including uncommitted work.

The principal crosswalk below is for provenance; definitions, claims, and their proof arguments appear in the manuscript itself. Links are relative to the YM directory containing the PDF and sources.

Manuscript material	Investigation record
Scope and global target	Assumed OS positivity and global scope; global source and Ward audit.
Fixed boundary and hierarchy	Preliminary positive hierarchy; finite-slab reflection and discrete bounds; first interacting test.
Winding and operator choices	Native winding/electric sources; topology and character options.
Character and gamma limits	Character channels and prime mixed pairings; electric parity and archimedean response.
Occurrence obstructions	Bounded winding occurrence; native recentering and coercive positive completion.
One detecting response	Single-probe occurrence, full current spectrum, and prime boundaries.
Determination and domains	Off-diagonal rigidity and compact existence; source domains and weight transport.
Spectral comparisons	Native generators and generalized sources; source spectra, modular and Liouville controls; primary-source verification ledger.

The audits are the [global-mechanisms review](#), [off-diagonal/domain review](#), [character, current, and prime audit](#), [electric-parity review](#), [OS and source-domain audit](#), and [single-probe and cutoff audit](#). They are model-assisted assessments; their names do not certify independent human review. The [consolidation audit](#) records the editorial checks and the corrected missing addition sign in (A.9).

C.2 Numerical evidence ledger

All the following are floating diagnostics, with small records retained under `numerics/records/`. They neither prove an infinite limiting statement nor certify positive completion.

Record family	Controls	Main scope
Positive hierarchy (24 September)	64	Finite algebra and prescribed smooth back-grounds
Finite slab (24 September)	—	Interacting experiment; includes 30 independent dense-unitary algebra tests
Winding/electric pairing (25 September)	64	Laurent and electric Gram formulas
Character channels (25 September)	57	Weighted adjoints, phases, finite-scale limits
Archimedean response (25 September)	19	Parity, contact, growth, phase failures
Electric-parity review (25 September)	76	Rerun of the preceding 57+19, not new distinct controls
Current recentering (25 September)	19	Sine scaling, retained and escaped norms
Single probe and prime edges (25 September)	13	Product conventions and sharp-cutoff profiles

The saved records report all listed controls passing in their recorded environments. They were not rerun merely to consolidate this manuscript. Several use Haar and the prescribed density $\rho_{\text{test}} = (1 + 0.35 \cos \theta + 0.2 \cos 2\theta)/0.9$; these are not interacting four-dimensional samples. Only the finite-slab experiment in Appendix A samples that Wilson action.

For the latest edge check the cutoffs were $10^3, 10^4, 10^5, 10^6$, with relative profile errors approximately 0.148035, 0.050738, 0.016088, 0.006306. The final two-edge energy ratio was approximately 0.999265. Refinement from 240 to 360 quadrature points changed the normalized energy by $8.204 \cdot 10^{-10}$. An initial 160/240 comparison missed the 10^{-7} refinement threshold at $1.0323 \cdot 10^{-7}$; quadrature was refined rather than the threshold loosened. Finite products with 28 and 40 factors check conventions, not the zero-free strip proof. No zeta zero list was used.

The [numerics index](#) provides the unchanged source programs and reproduction commands. Intermediate arrays remain in memory or external scratch space. The repository contains no new large datasets or third-party papers. Source hashes, build tools, page count, and PDF checks are recorded in `BUILD_RECORD.json`; instructions are in `BUILD.md`. Build intermediates are excluded from the maintained manuscript files.

C.3 Authorship and LLM assistance

Drafted for Edward Baker. Substantial assistance from an OpenAI large language model was used to develop and audit the investigation’s arguments, write diagnostic programs, and consolidate and typeset this manuscript. The model identity exposed in this session is **GPT-6 (Codex)**. The exact deployed variant and configured reasoning effort are **unavailable** and have not been inferred. These declarations do not imply author or independent specialist endorsement of every mathematical deduction. The proof arguments and their dependencies are supplied for review, and the manuscript’s conclusions retain the unresolved physical occurrence gap.

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